



US012406888B2

(12) **United States Patent**
Won et al.

(10) **Patent No.:** **US 12,406,888 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SEMICONDUCTOR SUBSTRATE AND METHOD OF DICING THE SAME**

(56) **References Cited**

U.S. PATENT DOCUMENTS

(71) Applicant: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)

7,382,038 B2 6/2008 Wu
8,076,756 B2 12/2011 Lane et al.
(Continued)

(72) Inventors: **Donghoon Won**, Cheonan-si (KR);
Jaeeun Lee, Suwon-si (KR);
Yeongkwon Ko, Suwon-si (KR);
Junyeong Heo, Suwon-si (KR)

FOREIGN PATENT DOCUMENTS

CN 101297394 A 10/2008
DE 10 2018 206 303 A1 10/2018
(Continued)

(73) Assignee: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)

OTHER PUBLICATIONS

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 692 days.

Notice of Allowance issued Jun. 29, 2025 by the Korean Patent Office for KR Patent Application No. 10-2019-0100527.
(Continued)

(21) Appl. No.: **16/871,189**

Primary Examiner — Erik T. K. Peterson

(22) Filed: **May 11, 2020**

(74) *Attorney, Agent, or Firm* — Sughrue Mion, PLLC

(65) **Prior Publication Data**

US 2021/0050264 A1 Feb. 18, 2021

(30) **Foreign Application Priority Data**

Aug. 16, 2019 (KR) 10-2019-0100527

(51) **Int. Cl.**

H01L 21/78 (2006.01)

H01L 21/268 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **H01L 21/78** (2013.01); **H01L 21/268** (2013.01); **H01L 23/562** (2013.01); **H01L 23/585** (2013.01)

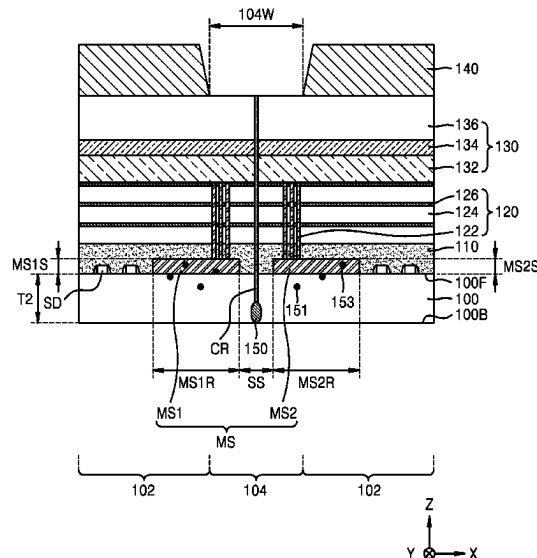
(58) **Field of Classification Search**

CPC H01L 21/78; H01L 23/585; B23K 26/53
See application file for complete search history.

(57) **ABSTRACT**

There is provided a method of dicing a semiconductor wafer, which includes providing a semiconductor substrate having a plurality of integrated circuit regions on an active surface of the semiconductor substrate, a dicing regions provided between adjacent integrated circuit regions of the plurality of integrated circuit regions, and a metal shield layer provided on the active surface across at least a portion of the adjacent integrated circuit regions and the dicing region, forming a modified layer by irradiating laser to an inside of the semiconductor substrate along the dicing region, propagating a crack from the modified layer in a direction perpendicular to a major-axial direction of the metal shield layer by polishing an inactive surface opposing the active surface of the semiconductor substrate and forming semiconductor chips by separating the adjacent integrated circuit regions, respectively, based on the crack propagating from the modified layer.

16 Claims, 20 Drawing Sheets



(51)	Int. Cl.		2010/0314619 A1 *	12/2010	Kaltalioglu	H01L 22/34
	H01L 23/00					257/E21.585
	H01L 23/58		2011/0241164 A1 *	10/2011	Nakamura	H01L 23/585
(56)	References Cited					257/774
			2012/0038028 A1 *	2/2012	Yaung	H01L 21/78
						257/E23.18
	U.S. PATENT DOCUMENTS		2012/0211748 A1 *	8/2012	Miccoli	H01L 21/78
						257/618
	8,299,580 B2	10/2012 Kumakawa et al.	2013/0075869 A1 *	3/2013	Mackh	H01L 21/78
	8,492,676 B2	7/2013 Azuma et al.				257/E23.179
	9,041,162 B2	5/2015 Miccoli et al.	2014/0264767 A1 *	9/2014	Gratz	H01L 23/585
	9,406,625 B2	8/2016 Wang et al.				257/620
	9,640,420 B2	5/2017 Nakamura	2016/0005697 A1 *	1/2016	Noh	H01L 23/562
	10,297,710 B2	5/2019 Kiriha				257/508
	10,418,335 B2	9/2019 Cho et al.	2016/0013138 A1 *	1/2016	Chen	H01L 21/565
	2002/0024115 A1 *	2/2002 Ibnabdeljalil				257/774
			2016/0163916 A1 *	6/2016	Ilievski	H01L 33/007
	2003/0122220 A1 *	7/2003 West				438/33
			2016/0380605 A1	12/2016	Matsumoto et al.	
	2005/0272223 A1 *	12/2005 Fujii	2017/0047221 A1 *	2/2017	Harada	H01L 21/67092
			2017/0207181 A1 *	7/2017	Nakamura	H01L 21/6836
	2006/0022224 A1 *	2/2006 Hiroi	2018/0175006 A1 *	6/2018	Yan	H01L 23/585
			2018/0261555 A1 *	9/2018	Cho	H01L 21/02675
	2006/0163699 A1 *	7/2006 Kumakawa	2019/0157150 A1 *	5/2019	Park	H01L 21/76811
			2020/0058551 A1 *	2/2020	Bae	H01L 21/6836
	2007/0066044 A1	3/2007 Abe et al.	2021/0098392 A1 *	4/2021	Wu	H01L 27/14634
	2007/0111477 A1	5/2007 Maruyama et al.				
	2008/0067690 A1 *	3/2008 Kumagai				
	2008/0203538 A1	8/2008 Kumakawa et al.				
	2009/0014882 A1 *	1/2009 Ito				
	2009/0121313 A1 *	5/2009 Hashimoto				
	2009/0121321 A1 *	5/2009 Miccoli				
	2009/0121337 A1 *	5/2009 Abe				
	2009/0201043 A1 *	8/2009 Kaltalioglu				
	2009/0321890 A1 *	12/2009 Jeng				
	2010/0123219 A1 *	5/2010 Chen				
	2010/0133659 A1 *	6/2010 Hara				
	2010/0283128 A1 *	11/2010 Chen				

FOREIGN PATENT DOCUMENTS

JP	2005-340423 A	12/2005
JP	2011-166183 A	8/2011
JP	2016-54204 A	4/2016
JP	2016-207765 A	12/2016
KR	10-2006-0020627 A	3/2006
KR	10-2006-0085165 A	7/2006
KR	10-2010-0036241 A	4/2010
KR	10-2017-0053112 A	5/2017
KR	10-2018-0104261 A	9/2018

OTHER PUBLICATIONS

Communication issued on Jul. 7, 2025 by the China National Intellectual Property Administration in Chinese Patent Application No. 202010673029.1.

* cited by examiner

FIG. 1

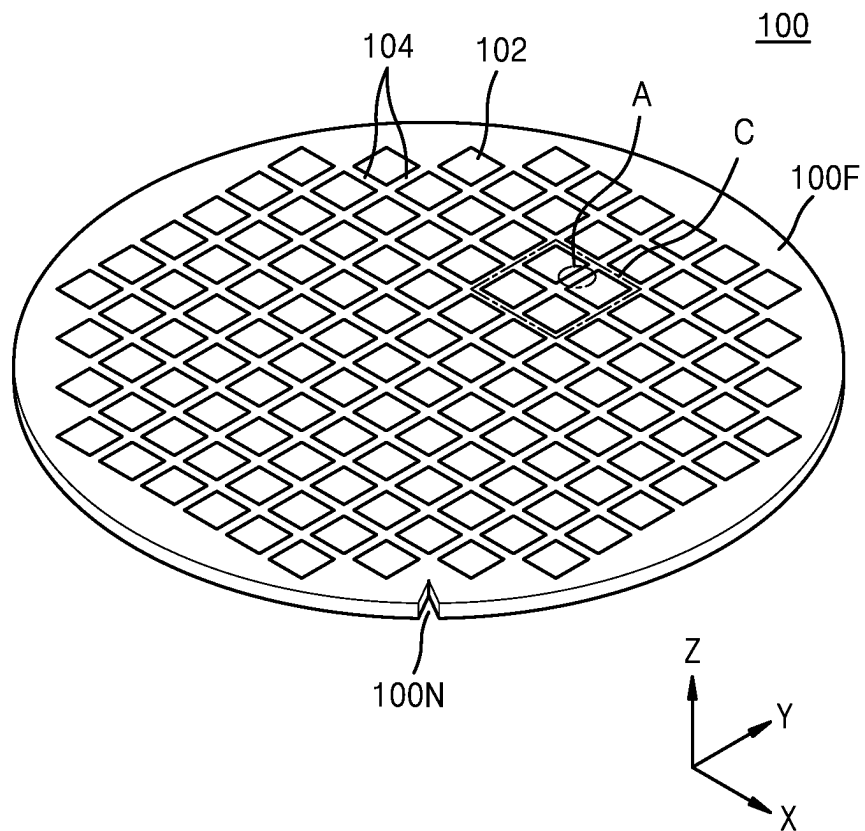


FIG. 2A

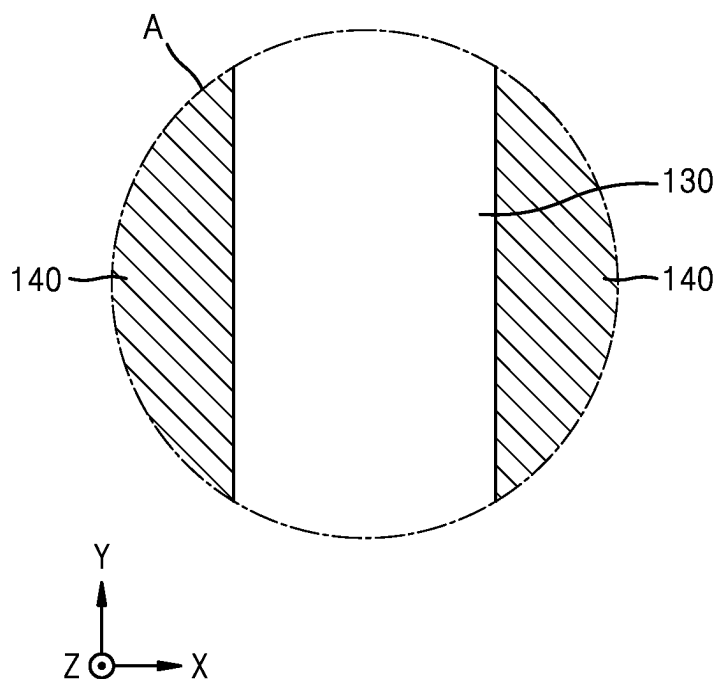


FIG. 2B

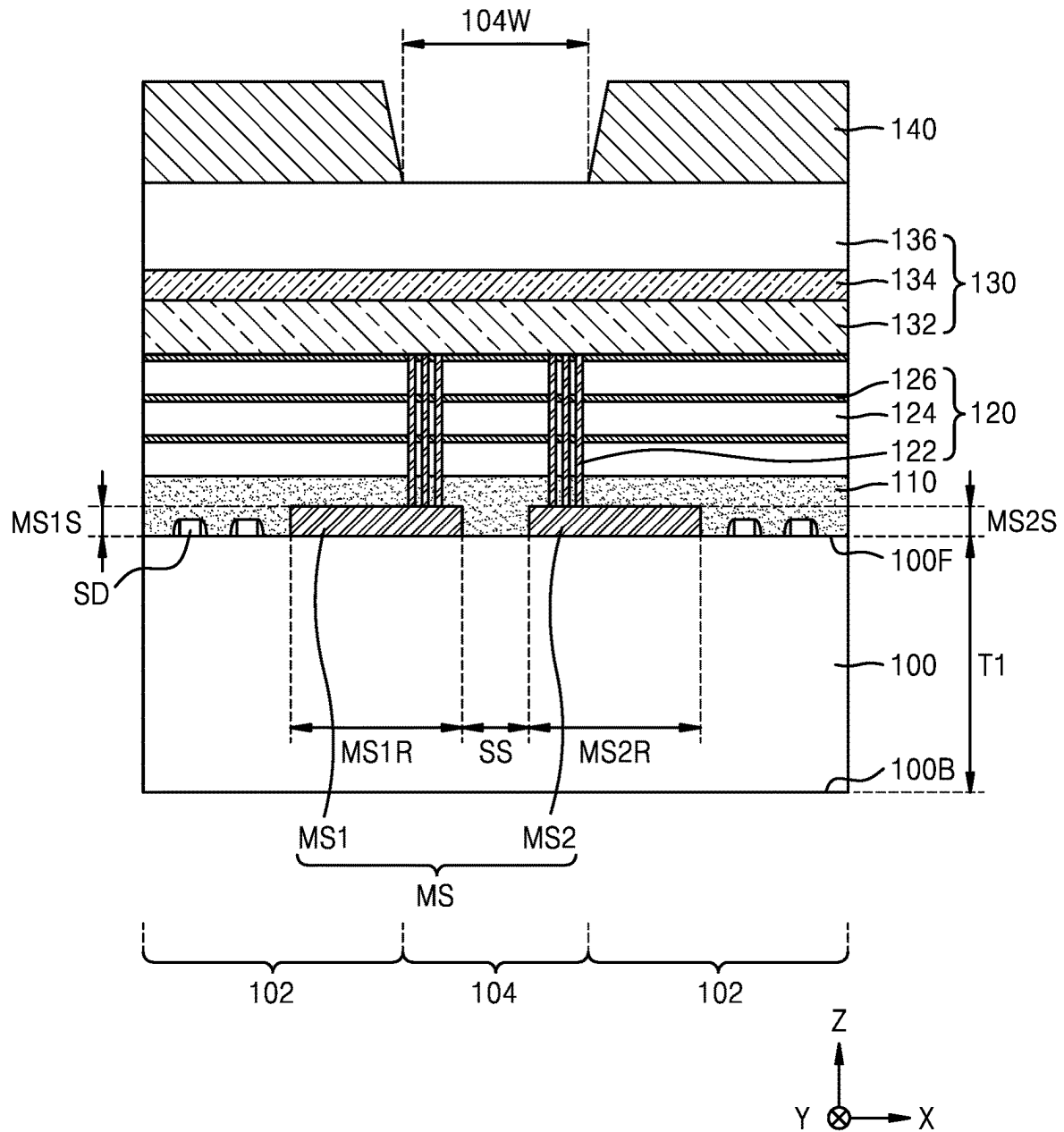


FIG. 2C

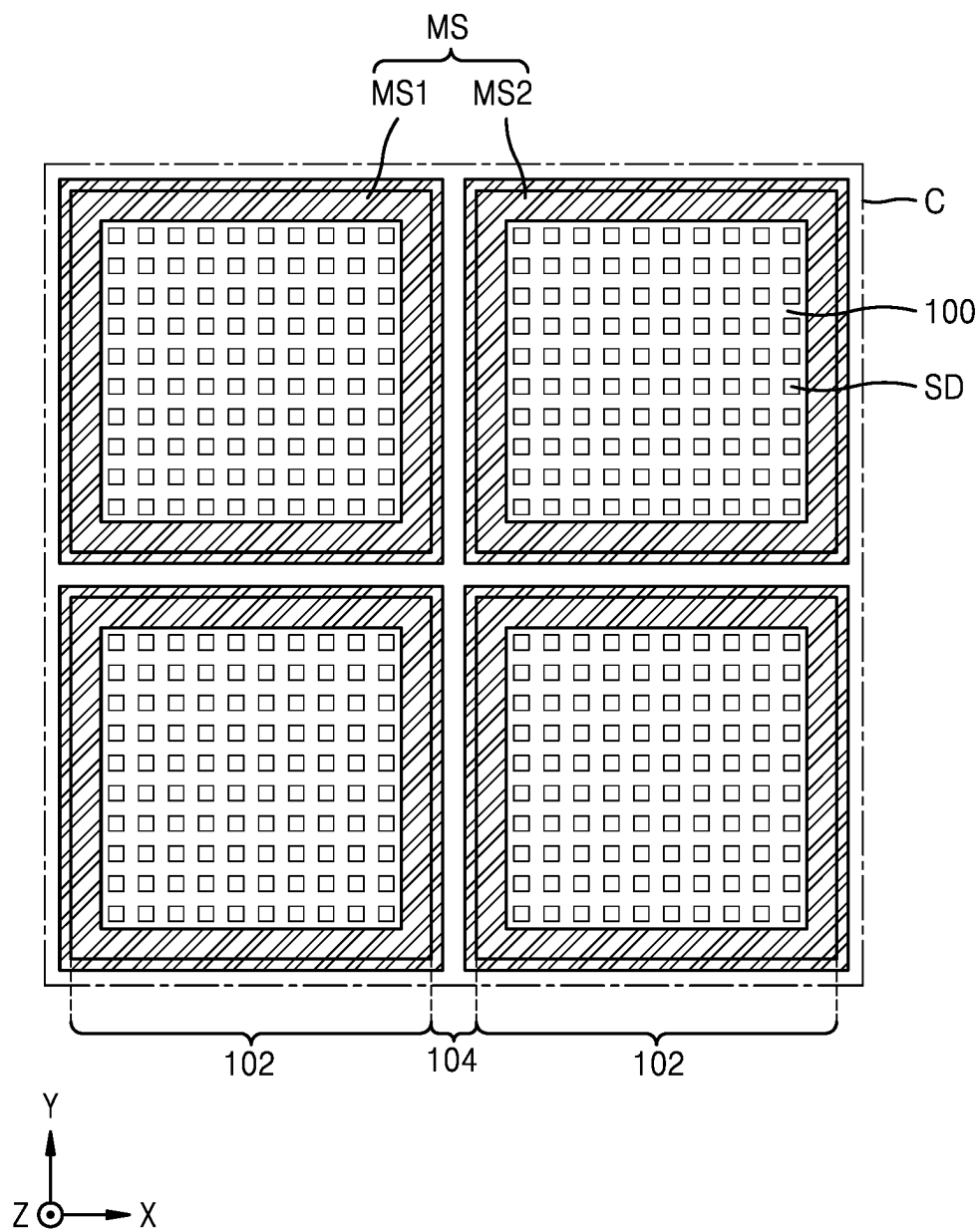


FIG. 3

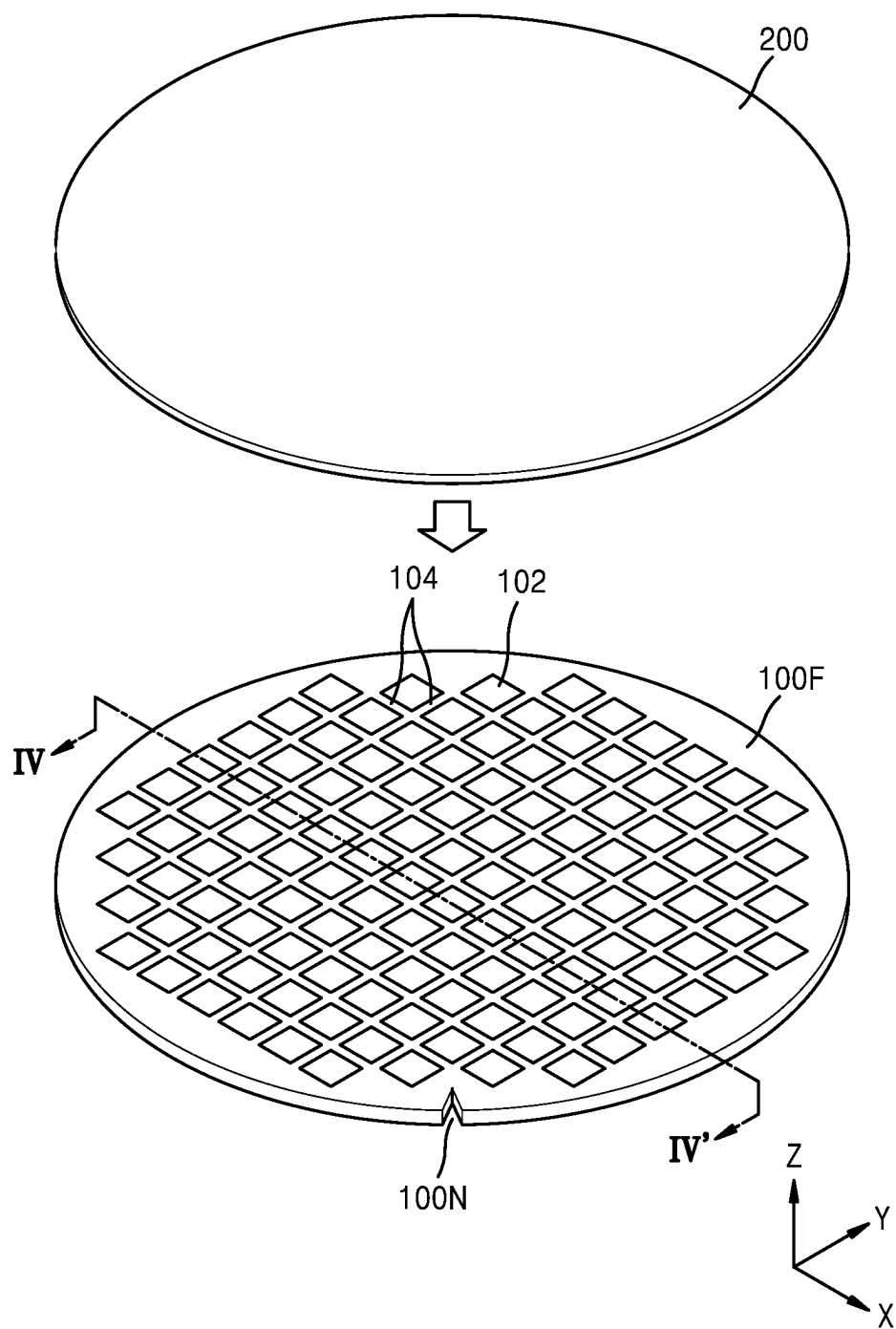


FIG. 4

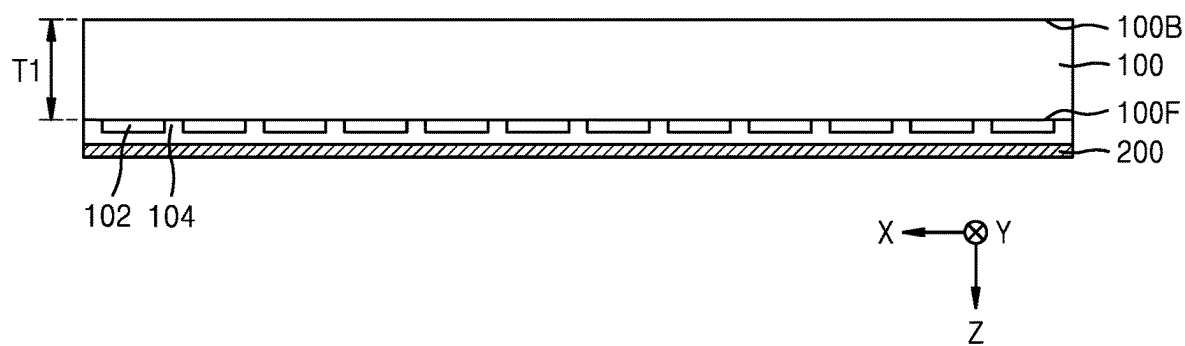


FIG. 5A

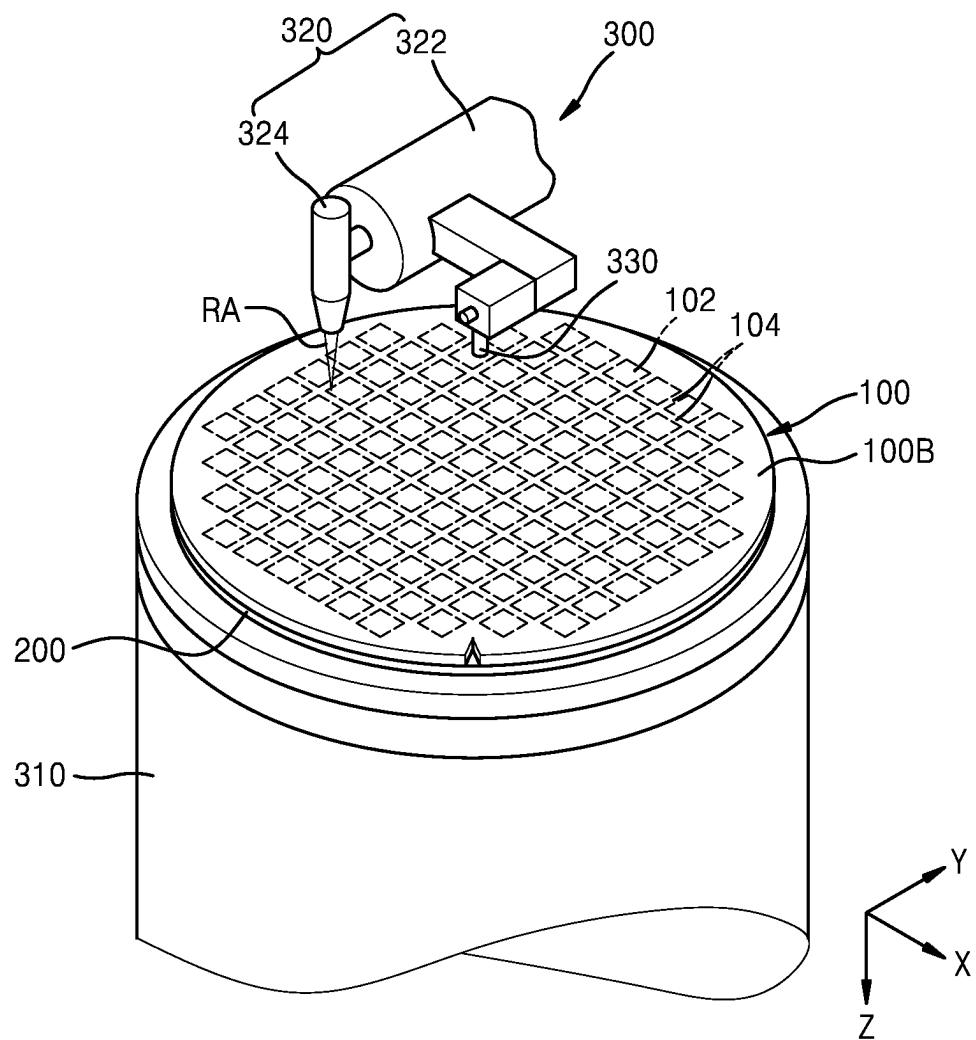


FIG. 5B

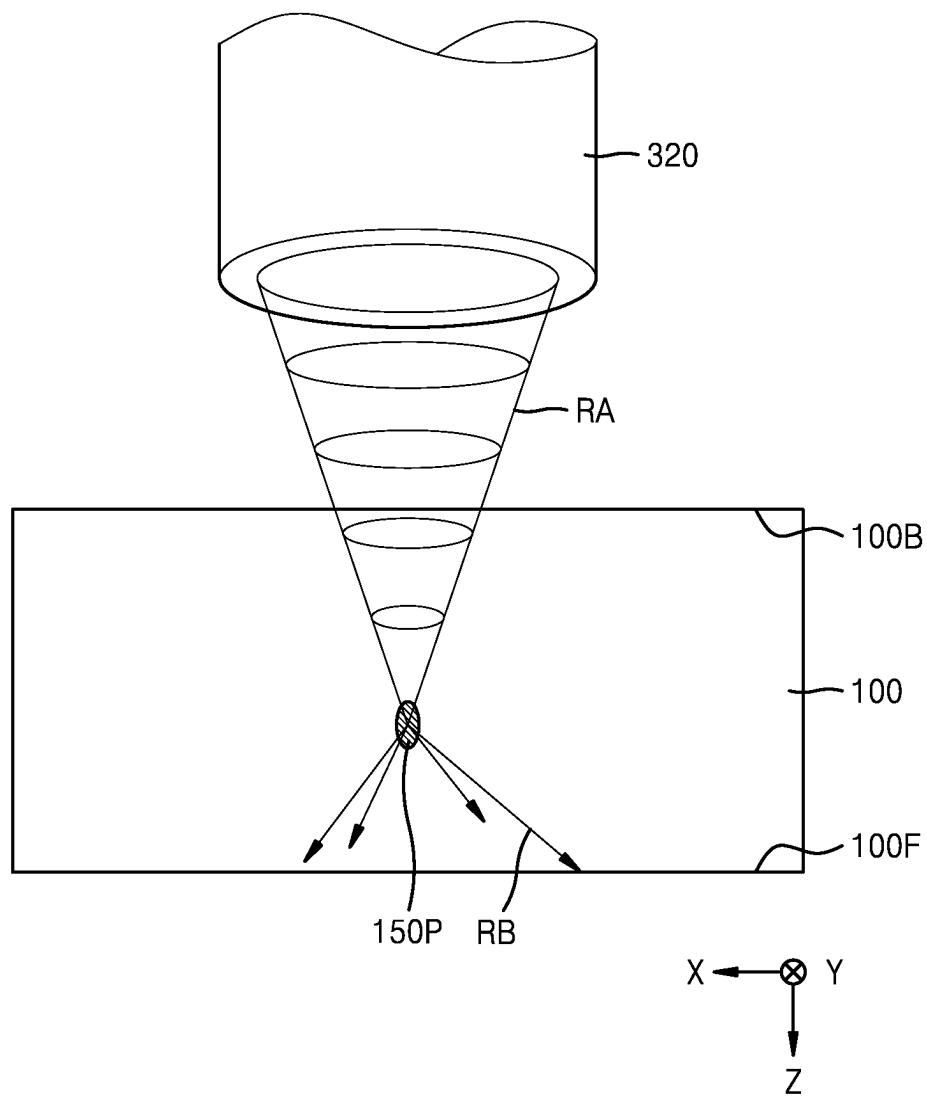


FIG. 6

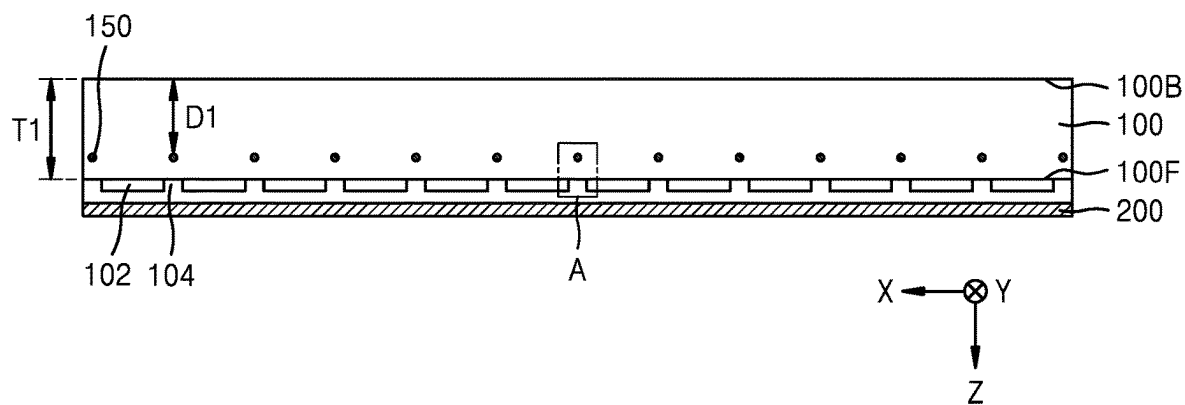


FIG. 7A

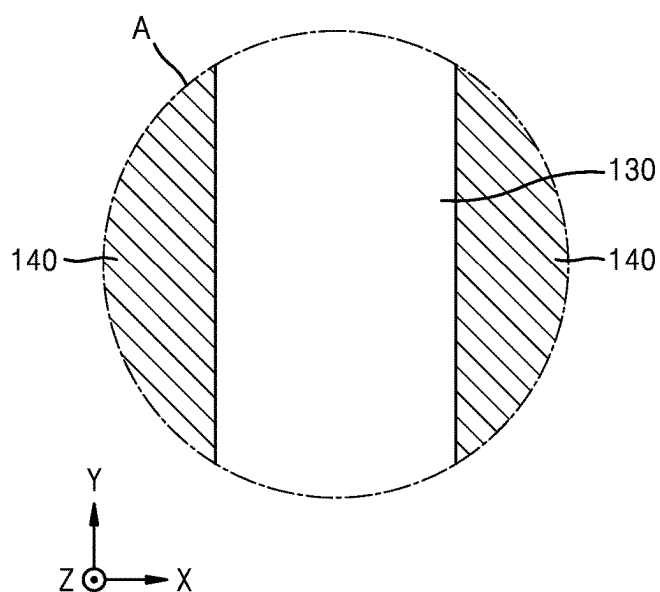


FIG. 8

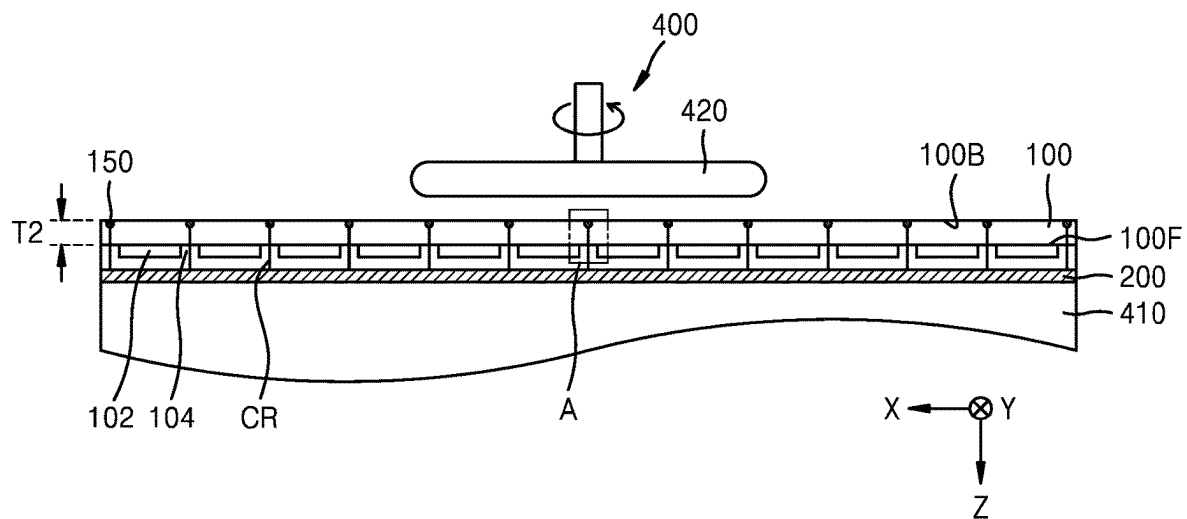


FIG. 9A

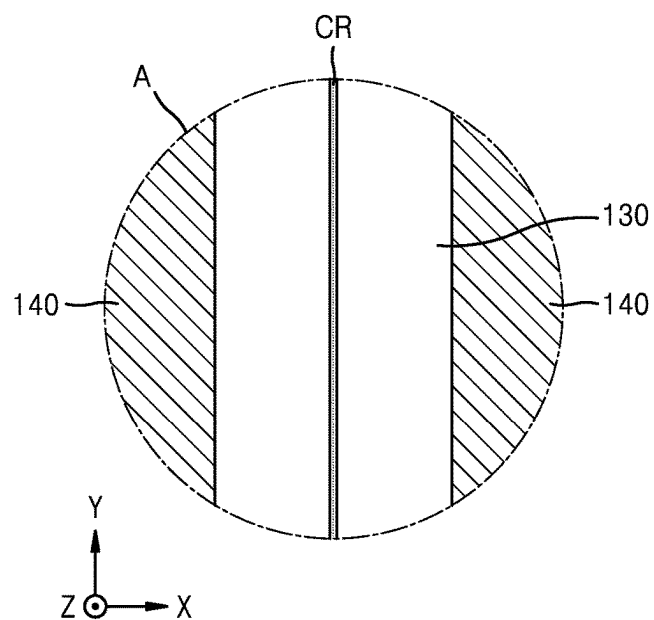


FIG. 9B

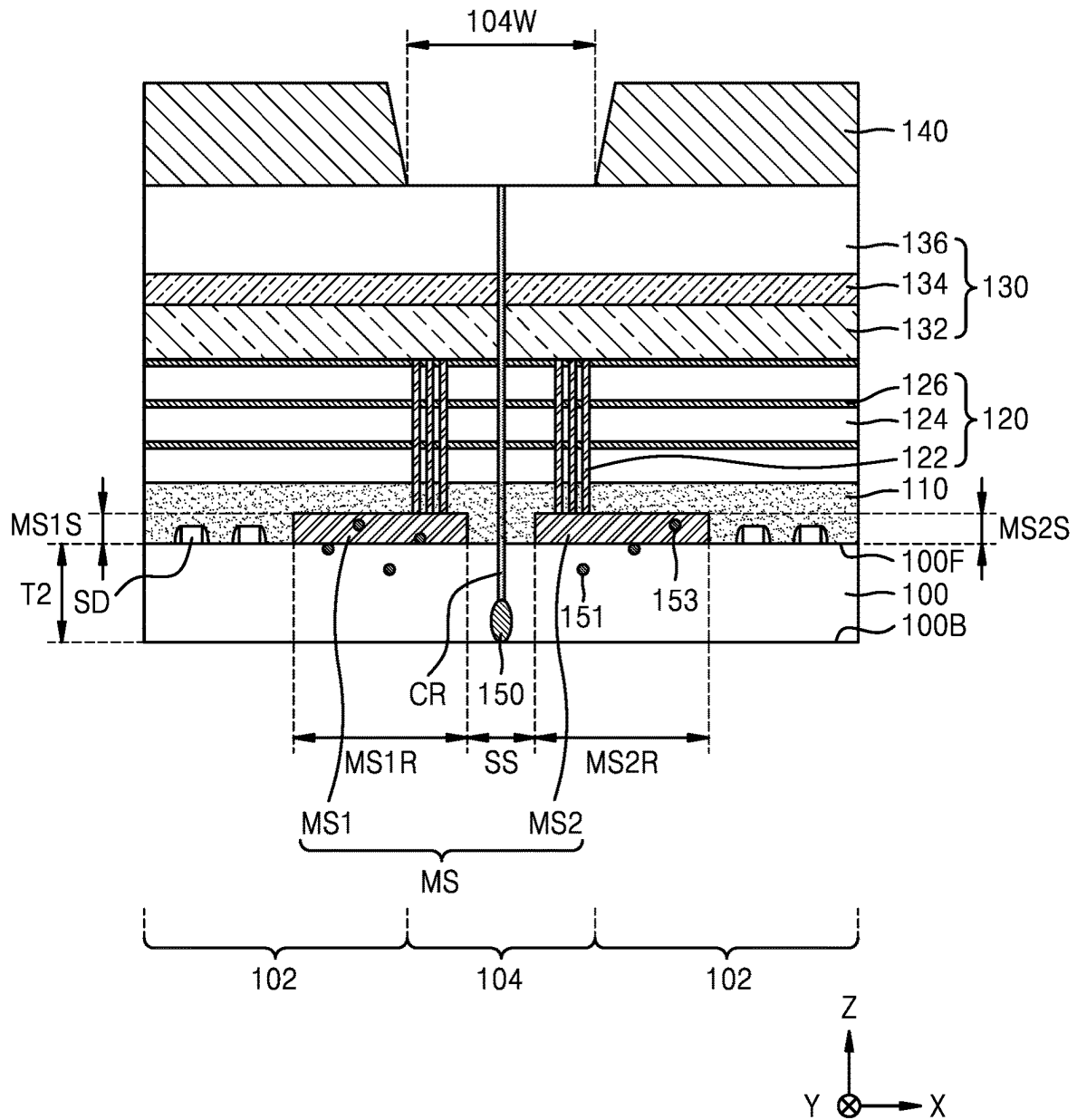


FIG. 10

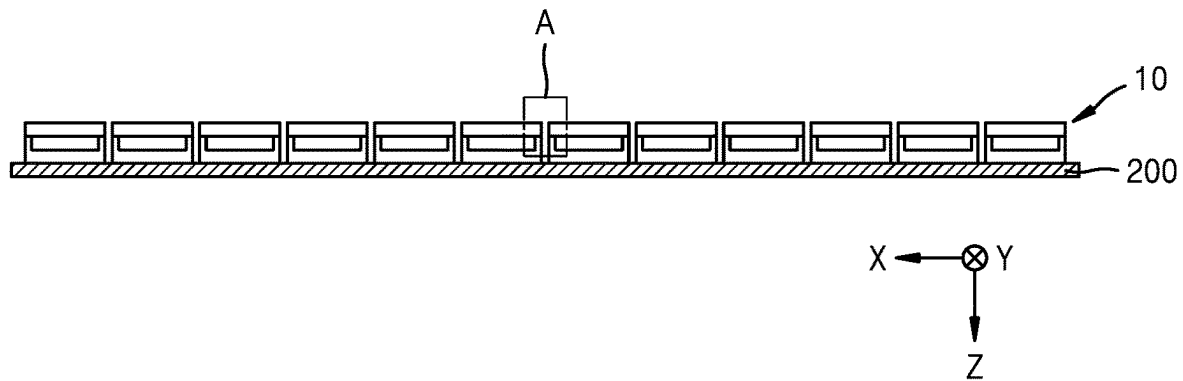
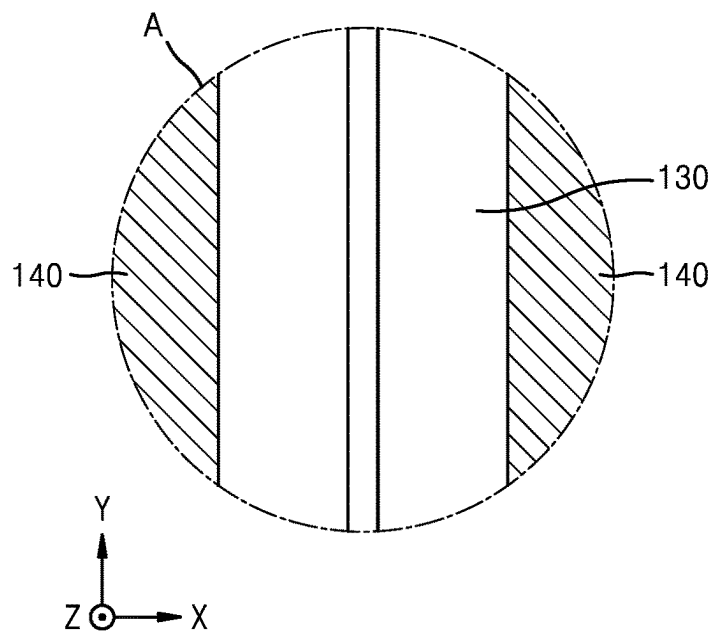


FIG. 11A



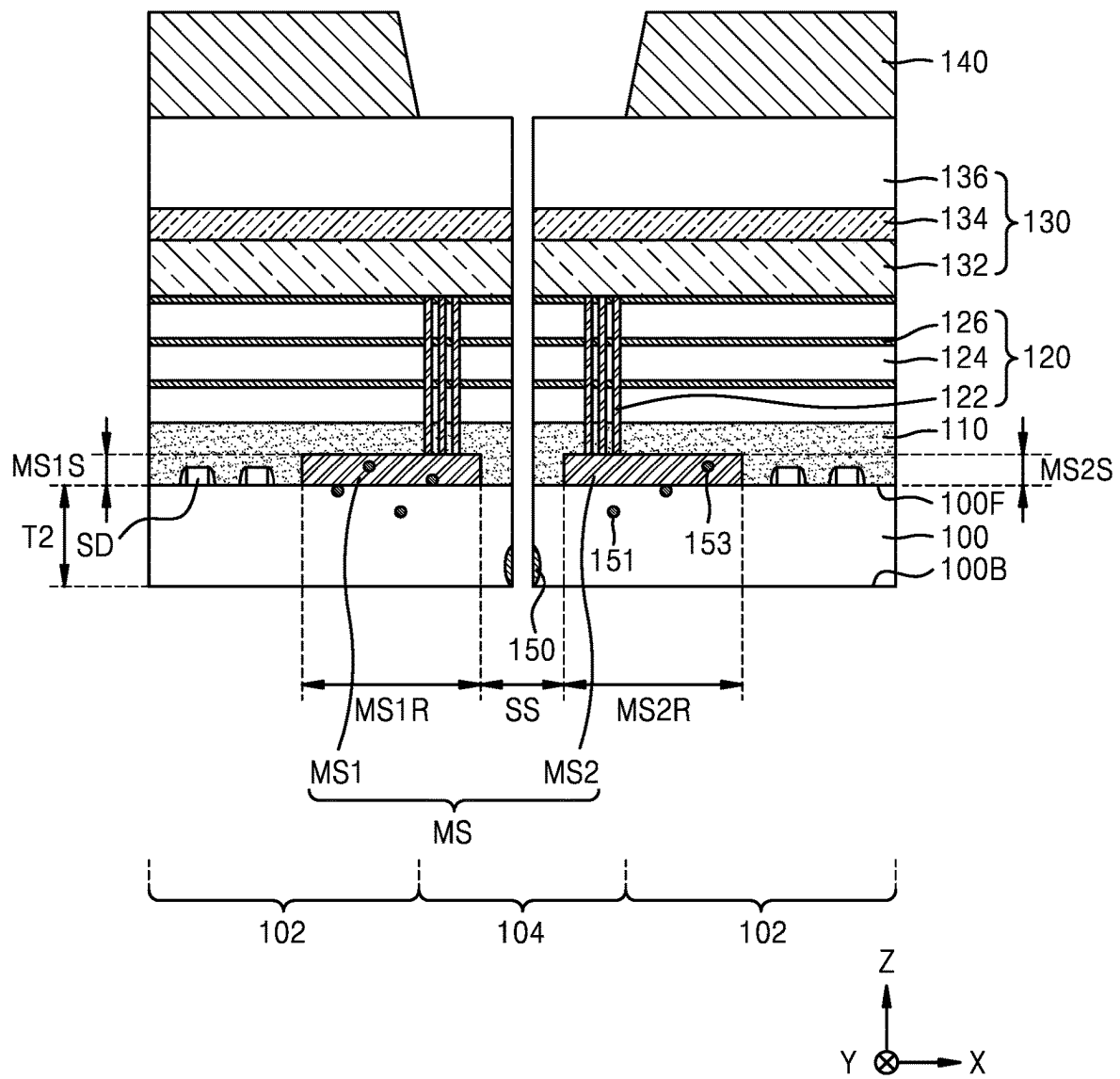


FIG. 12A

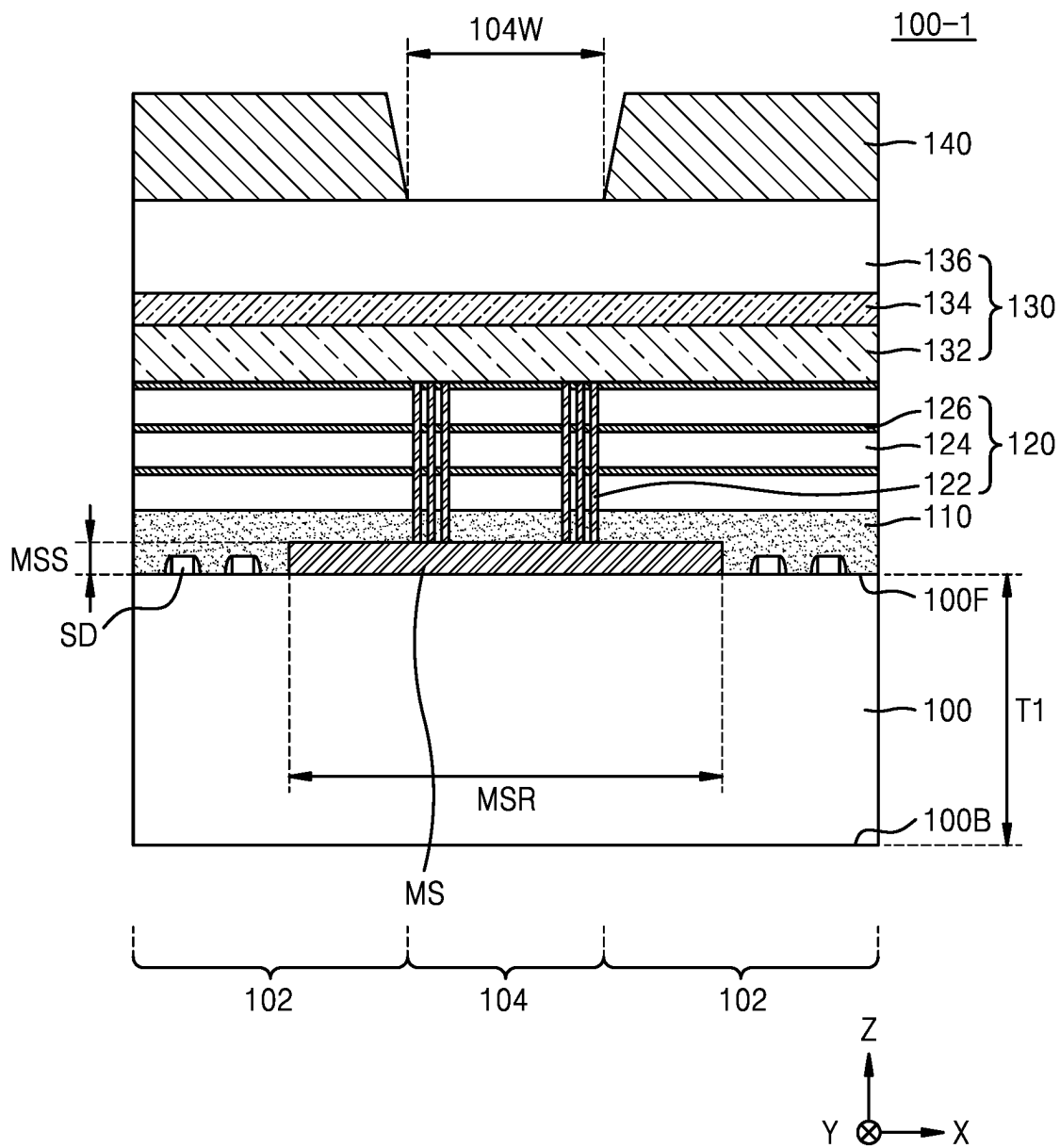


FIG. 12B

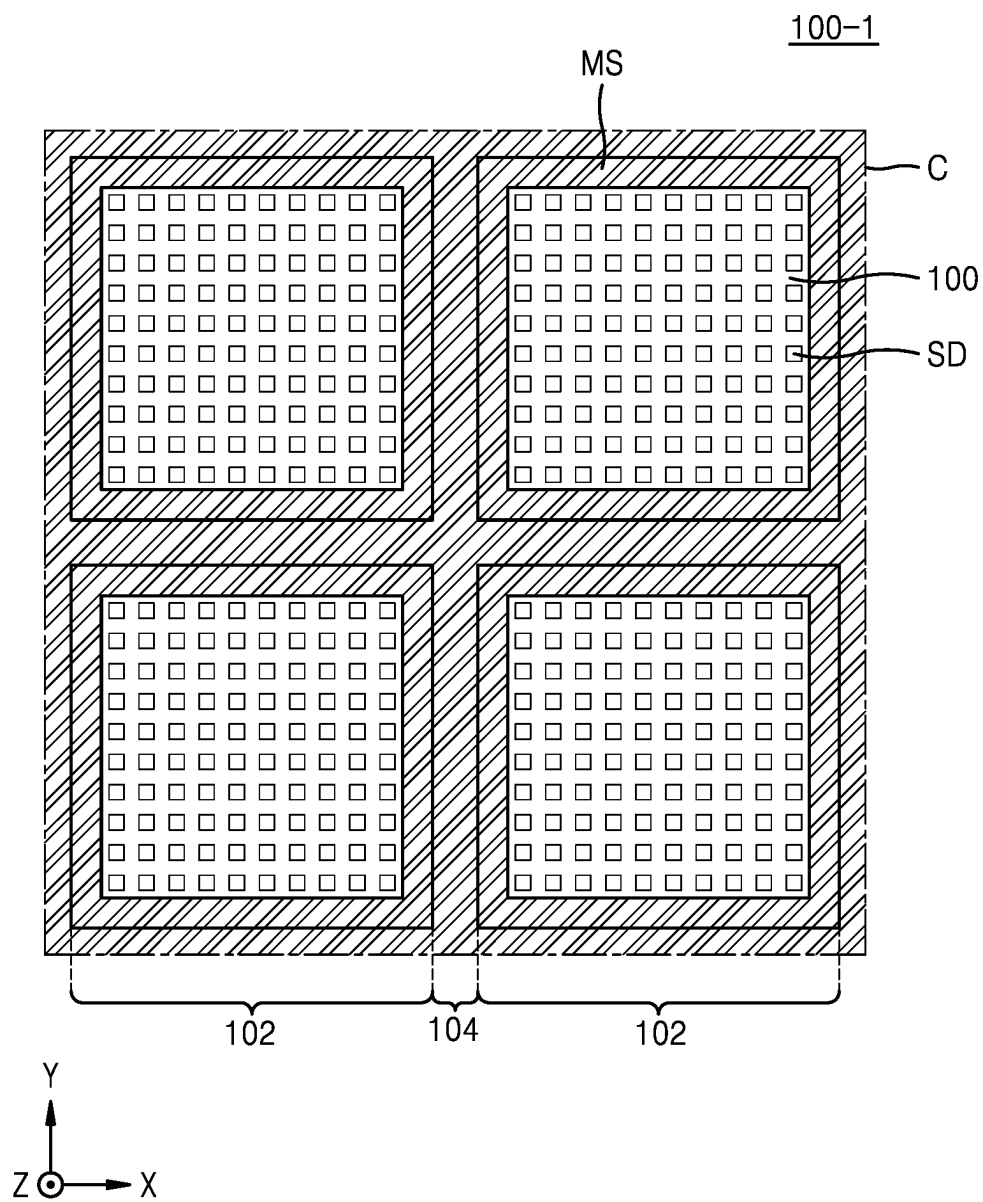


FIG. 13

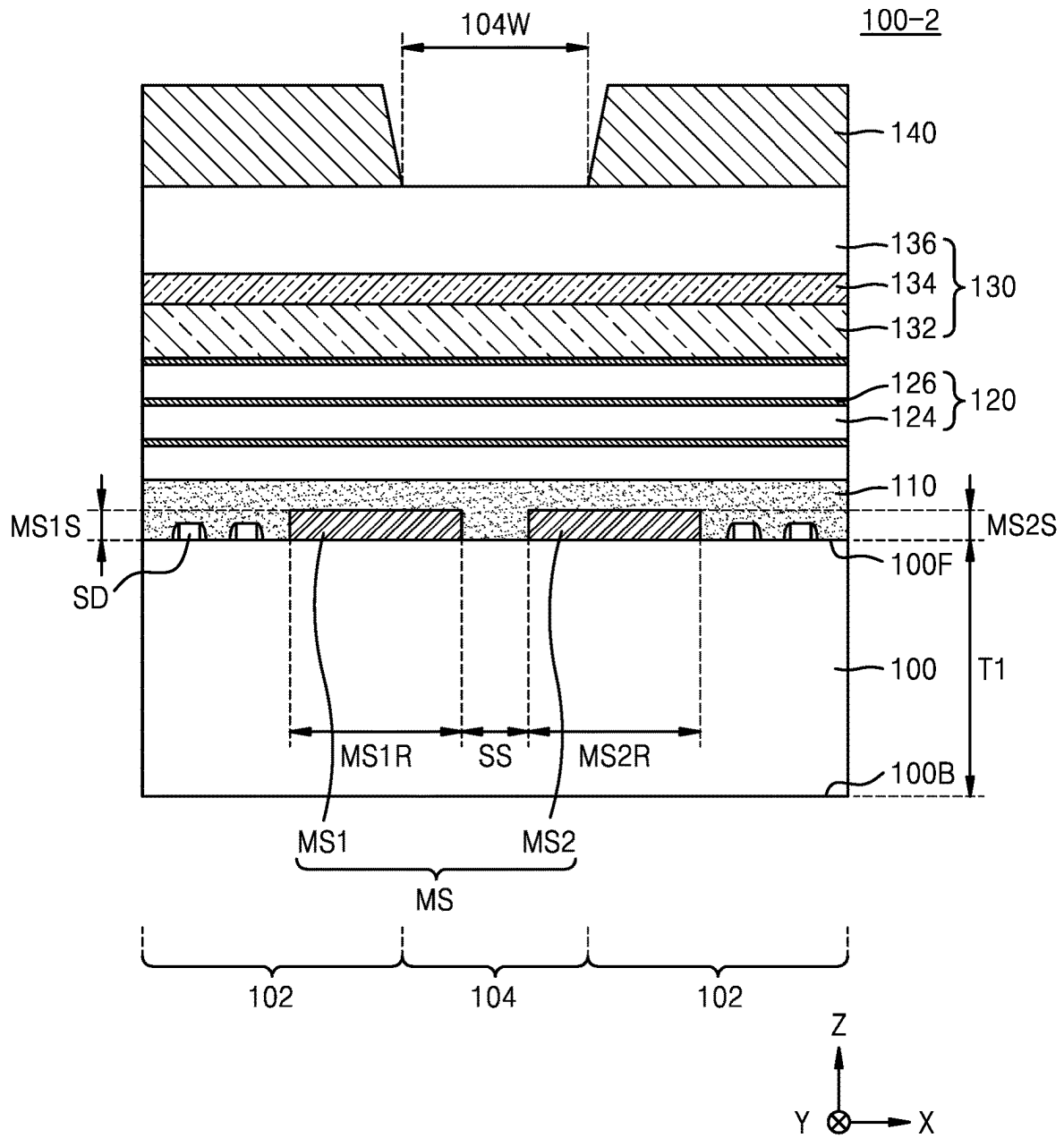


FIG. 14

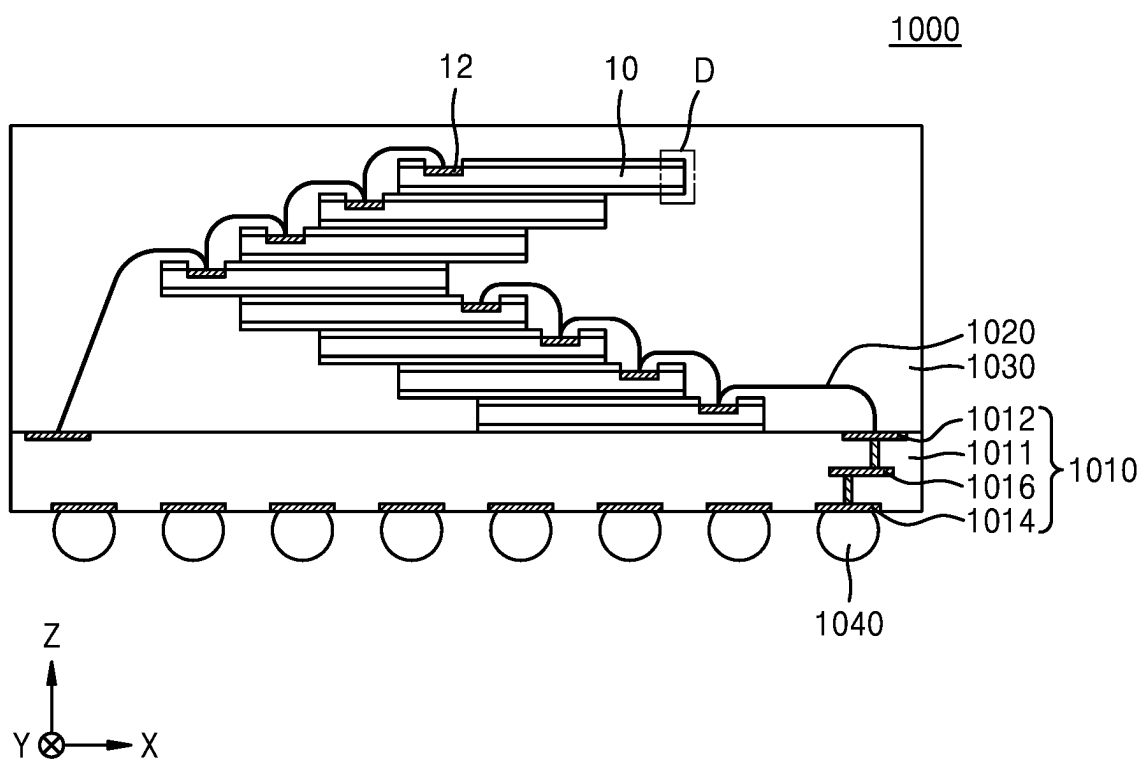


FIG. 15B

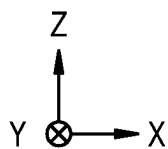
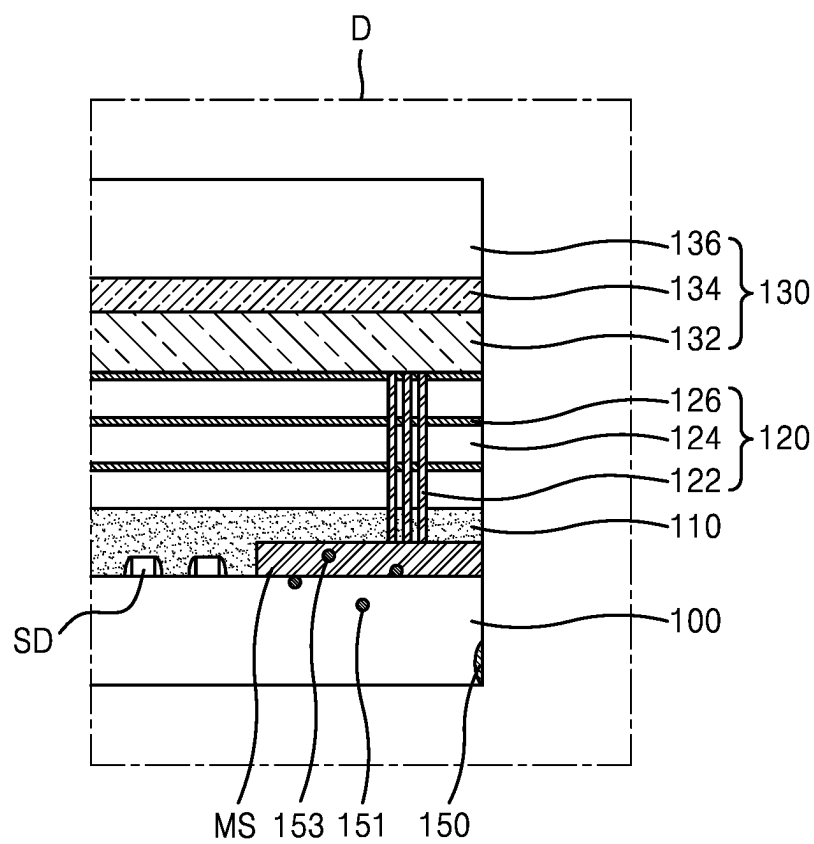
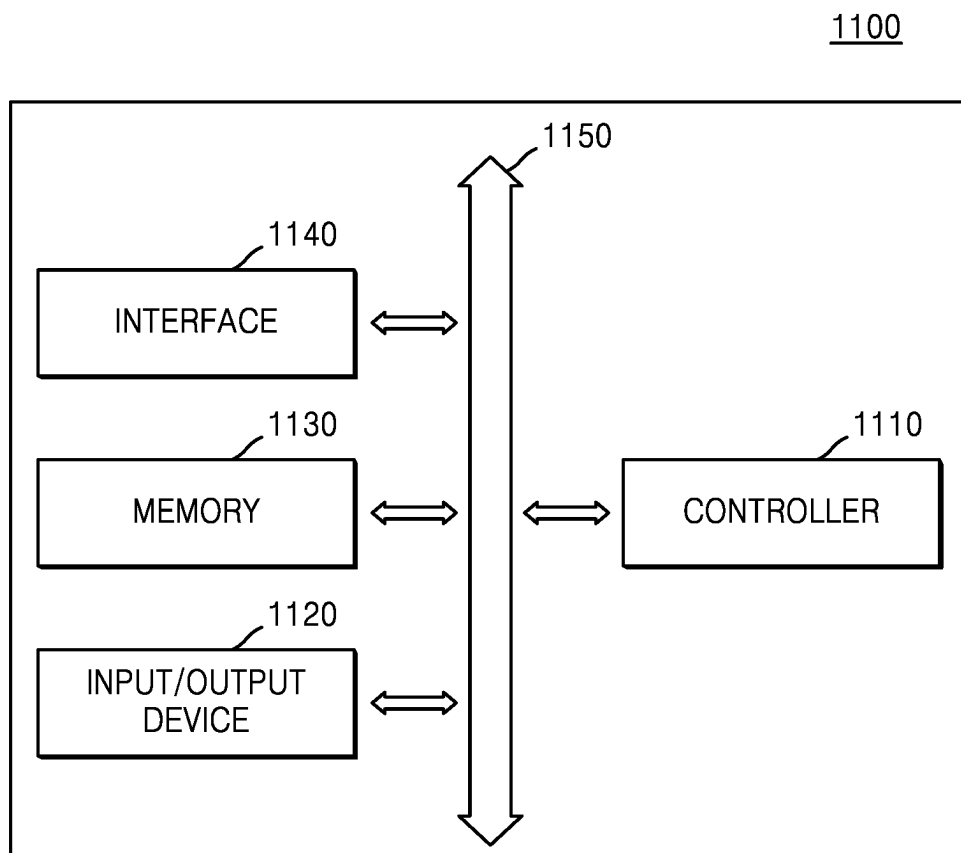


FIG. 16



1

SEMICONDUCTOR SUBSTRATE AND METHOD OF DICING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application is based on and claims priority from Korean Patent Application No. 10-2019-0100527, filed on Aug. 16, 2019, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND

The disclosure relates to a semiconductor substrate and a method of dicing the same, and more particularly, to a method of cutting a semiconductor substrate by using laser.

In a chip manufacturing process, after integrated circuits are formed on an active surface of a semiconductor substrate, an inactive surface of the semiconductor substrate is polished and the polished semiconductor substrate is diced to separate the integrated circuits into respective semiconductor chips. Generally, the polished semiconductor substrate is mechanically diced using a sawing blade. When mechanical dicing is performed in this way, a diced surface of the semiconductor chips may break, causing many faults in the semiconductor chips. Hence, a method of dicing a semiconductor substrate using laser has been studied.

SUMMARY

The disclosure provides a semiconductor substrate that suppresses faults generation and a method of dicing the semiconductor substrate, in a process of dicing the semiconductor substrate into semiconductor chips by using laser.

Aspects of the disclosure are not limited to the foregoing, and other unmentioned aspects would be apparent to one of ordinary skill in the art from the following description.

According to an aspect of the disclosure, there is provided a method of dicing a semiconductor wafer, the method comprising: providing a semiconductor substrate having a plurality of integrated circuit regions on an active surface of the semiconductor substrate, a dicing regions provided between adjacent integrated circuit regions of the plurality of integrated circuit regions, and a metal shield layer provided on the active surface across at least a portion of the adjacent integrated circuit regions and the dicing region; forming a modified layer by irradiating laser to an inside of the semiconductor substrate along the dicing region; propagating a crack from the modified layer in a direction perpendicular to a major-axial direction of the metal shield layer by polishing an inactive surface opposing the active surface of the semiconductor substrate; and forming semiconductor chips by separating the adjacent integrated circuit regions, respectively, based on the crack propagating from the modified layer.

According to another aspect of the disclosure, there is provided a method of dicing a semiconductor wafer, the method comprising: providing a semiconductor substrate having a plurality of integrated circuit regions on an active surface of the semiconductor substrate, a dicing regions provided between adjacent integrated circuit regions of the plurality of integrated circuit regions, and a metal shield layer formed on the active surface across at least a portion of the integrated circuit regions and the dicing region; forming a modified layer by irradiating laser to an inside of the semiconductor substrate along the dicing region; propa-

2

gating a crack from the modified layer in a direction perpendicular to a major-axial direction of the metal shield layer by polishing an inactive surface opposing the active surface of the semiconductor substrate; and forming semiconductor chips by separating the adjacent integrated circuit regions, respectively, based on the crack propagating from the modified layer, wherein, in a cross-sectional view, the metal shield layer includes a first metal shield layer and a second metal shield layer with a space region therebetween in a location where the crack propagates, each of the first metal shield layer and the second metal shield layer includes a major axis that is parallel to the active surface and a minor axis that is perpendicular to the active surface in the cross-sectional view, and a length of the major axis is about 50 μm to about 100 μm , and a length of the minor axis is about 0.5 μm to about 1 μm .

According to an aspect of the disclosure, there is provided a semiconductor wafer comprising: a semiconductor substrate comprising a plurality of integrated circuit regions on an active surface; a dicing regions provided between adjacent integrated circuit regions of the plurality of integrated circuit regions; and a metal shield layer provided on the active surface across a portion of the adjacent integrated circuit regions and the dicing region, wherein, in a cross-sectional view, the metal shield layer comprises a major axis that is parallel to the active surface and a minor axis that is perpendicular to the active surface, and in a planar view, the metal shield layer is arranged along a circumference of a respective integrated circuit region, among the adjacent integrated circuit regions.

According to an aspect of the disclosure, there is provided a semiconductor device comprising: a semiconductor substrate comprising an integrated circuit region and a dicing region on an active surface of the semiconductor substrate, the dicing region being provided adjacent to the integrated circuit region; a metal shield layer provided on the active surface across the integrated circuit region and the dicing region; a device layer comprising one or more semiconductor devices provided on the active surface of the semiconductor substrate; one or more wires provided in a wiring layer formed on the device layer; and one or more vertical metal structures formed on the metal shield layer, wherein, in a cross-sectional view, the metal shield layer comprises a major axis that is parallel to the active surface and a minor axis that is perpendicular to the active surface, and wherein, in the cross-sectional view, the one or more vertical metal structures are perpendicular to the major axis of the metal shield layer.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of the disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:

FIG. 1 is a perspective view of a semiconductor substrate according to an exemplary embodiment of the disclosure;

FIG. 2A is an enlarged plane view of a region A illustrated in FIG. 1;

FIG. 2B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. 2A in a direction X;

FIG. 2C is an enlarged plane view of a region C illustrated in FIG. 1;

FIG. 3 is a perspective view showing a state in which a protective sheet is attached to a semiconductor substrate, according to an embodiment of the disclosure;

3

FIG. 4 is a side cross-sectional view corresponding to a line IV-IV' illustrated in FIG. 3, where attachment of a protective sheet to a semiconductor substrate is completed;

FIG. 5A is a perspective view showing a state where laser is irradiated into a semiconductor substrate according to an embodiment of the disclosure;

FIG. 5B is a side cross-sectional view showing a traveling direction of laser;

FIG. 6 is a side cross-sectional view showing a state where irradiation of laser is completed, according to an embodiment of the disclosure;

FIG. 7A, which corresponds to FIG. 2A, is an enlarged plane view of a region A illustrated in FIG. 6;

FIG. 7B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. 7A in a direction X;

FIG. 8 is a side cross-sectional view showing a state where a semiconductor substrate according to an embodiment of the disclosure is polished;

FIG. 9A, which corresponds to FIG. 2A, is an enlarged plane view of a region A illustrated in FIG. 8;

FIG. 9B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. 9A in a direction X;

FIG. 10 is a side cross-sectional view showing a state where a semiconductor substrate according to an embodiment of the disclosure is diced into semiconductor chips;

FIG. 11A, which corresponds to FIG. 2A, is an enlarged plane view of a region A illustrated in FIG. 10;

FIG. 11B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. 11A in a direction X;

FIGS. 12A and 12B illustrate a semiconductor substrate according to another embodiment of the disclosure, in which FIG. 12A is an enlarged side cross-sectional view of a region A illustrated in FIG. 1 and FIG. 12B is an enlarged plane view of a region C illustrated in FIG. 1;

FIG. 13 illustrates a semiconductor substrate according to another embodiment of the disclosure, in which FIG. 13 is an enlarged side cross-sectional view of a region A illustrated in FIG. 1;

FIG. 14 is a side cross-sectional view illustrating a semiconductor package including semiconductor chips diced from a semiconductor substrate according to an embodiment of the disclosure;

FIG. 15A is an enlarged side cross-sectional view of a portion D illustrated in FIG. 14 according to an embodiment of the disclosure;

FIG. 15B is an enlarged side cross-sectional view of a portion D illustrated in FIG. 14 according to an embodiment of the disclosure;

FIG. 16 is a structural diagram showing a system of a semiconductor package including semiconductor chips diced from a semiconductor substrate according to an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

Hereinafter, embodiments of the disclosure will be described in detail with reference to the accompanying drawings.

FIG. 1 is a perspective view of a semiconductor substrate according to an exemplary embodiment of the disclosure, FIG. 2A is an enlarged plane view of a region A illustrated in FIG. 1, FIG. 2B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. 2A in a direction X, and FIG. 2C is an enlarged plane view of a region C illustrated in FIG. 1.

4

Referring to FIG. 1, FIGS. 2A, 2B and 2C, a semiconductor substrate 100 includes integrated circuit regions 102 and a dicing region 104.

According to an exemplary embodiment, a semiconductor substrate 100 may include a wafer and may have a circular shape having a constant first thickness T1. The semiconductor substrate 100 may have a notch 100N used as a reference point of wafer alignment.

The semiconductor substrate 100 may include, for example, silicon. Alternatively, the semiconductor substrate 100 may include a semiconductor element like germanium or a compound semiconductor such as silicon carbide (SiC), gallium arsenide (GaAs), indium arsenide (InAs), and indium phosphide (InP). The semiconductor substrate 100 may have a silicon-on-insulator (SOI) structure. In some embodiments of the disclosure, the semiconductor substrate 100 may include a well or structure doped with impurities, which is a conductive region. The semiconductor substrate 100 may have various element isolation structures such as a shallow trench isolation (STI) structure.

Herein, the semiconductor substrate 100 is assumed to have a diameter of about 12 inches, and a description will be made of a case where a silicon wafer is used. However, it will be understood by those of ordinary skill in the art that the semiconductor substrate 100 having a diameter greater or less than the above-described diameter may be used and the semiconductor substrate 100 formed of other materials may also be used. The semiconductor substrate 100 may have a first thickness T1 of about 0.1 mm to about 1 mm. When the first thickness T1 of the semiconductor substrate 100 is too small, mechanical strength may be unsatisfactory; when the first thickness T1 is too large, a time spent in subsequent grinding increases, degrading the productivity of a semiconductor chip.

The semiconductor substrate 100 may include an active surface 100F that is a front side and an inactive surface 100B that is a back side. According to an exemplary embodiment, multiple integrated circuit regions 102 may be formed on the active surface 100F, and the multiple integrated circuit regions 102 are to be separated into semiconductor chips (10 of FIG. 10).

According to an exemplary embodiment, a semiconductor device SD may be formed on the active surface 100F of the semiconductor substrate. According to an exemplary embodiment, the semiconductor substrate may be classified into a memory device and a logic device.

The memory device may include a volatile memory device or a non-volatile memory device. The volatile memory device may include an existing volatile memory device and a currently developed volatile memory device, for example, dynamic random access memory (DRAM), static RAM (SRAM), thyristor RAM (TRAM), zero capacitor RAM (ZRAM), or twin transistor RAM (TTRAM). The non-volatile memory device may include an existing non-volatile memory device and a currently developed non-volatile memory device, for example, flash memory, magnetic RAM (MRAM), spin-transfer torque (STT)-MRAM, ferroelectric RAM (FRAM), phase change RAM (PRAM), resistive RAM (RRAM), nanotube RRAM, polymer RAM, nano floating gate memory, holographic memory, molecular electronics memory, or insulator resistance change memory.

The logic device may be implemented with, but not limited to, a microprocessor, a graphics processor, a signal processor, a network processor, an audio codec, a video codec, an application processor, a system-on-chip, etc. The microprocessor may include, for example, a single core or a multi-core.

The integrated circuit regions **102** may be arranged to be isolated from one another by the dicing region **104**. The dicing region **104** may be referred to as a scribe lane. The dicing region **104** may extend in a form of a cross in a first direction X and a second direction Y that is perpendicular to the first direction X. The dicing region **104** may have a form of a linear lane having a constant width **104W**.

That is, the integrated circuit regions **102** may be arranged to be separated from one another by being surrounded by the dicing region **104** in all directions. As will be described below, as the semiconductor substrate **100** and various types of material layers formed on the semiconductor substrate **100** are diced through a dicing process performed along the dicing region **104**, the integrated circuit regions **102** may be separated into a plurality of semiconductor chips (**10** of FIG. **10**) from one another.

A semiconductor device layer **110** may be formed on the active surface **100F** of the semiconductor substrate **100**. The semiconductor device layer **110** may correspond to a plurality of semiconductor devices (SDs) in the integrated circuit regions **102** and may correspond to a region including a plurality of semiconductor dummy devices in the dicing region **104**.

Multi-layer wires **120** may be formed from a top surface of the semiconductor device layer **110** to a bottom surface of an upper material film **130**. The multi-layer wires **120** may include an inter-layer insulating film **124** and a metal wire **126** that are arranged alternately. The multi-layer wires **120** may include a plurality of metal vertical structures **122** arranged in a third direction Z that is perpendicular to the active surface **100F** of the semiconductor substrate **100**.

The metal wire **126** may include a conductive material including at least one of aluminum (Al), copper (Cu), nickel (Ni), tungsten (W), platinum (Pt), and gold (Au), and the plurality of metal vertical structures **122** may include a material that is substantially the same as the metal wire **126**.

The inter-layer insulating film **124** may include a low dielectric material. The low dielectric material, which has a lower dielectric constant than that of a silicon oxide, may be useful for high integration and high speed of the semiconductor device SD when the low dielectric material is used as the inter-layer insulating film **124** in the semiconductor device SD.

In some exemplary embodiments of the disclosure, the inter-layer insulating film **124** may be formed to have a structure in which a first inter-layer insulating film, a second inter-layer insulating film, and a third inter-layer insulating film are sequentially stacked on the metal wire **126**. However, the number of films of the inter-layer insulating film **124** is not limited to the above-described example. The inter-layer insulating film **124** may be formed to fill peripheries of the plurality of metal vertical structures **122** and the metal wire **126** that are formed of a conductive material.

According to an exemplary embodiment, opposite side walls of each of the plurality of metal vertical structures **122** may be formed flat. Each of the plurality of metal vertical structures **122** may be formed of single metal, may have flat side walls, and may have a bar shape with a major axis in the third direction Z and a minor axis in the first direction X. Thus, the plurality of metal vertical structures **122** do not include a bonding interface and a crack producing portion between heterogeneous materials, thereby properly blocking the propagation of a crack described below (CR of FIG. **9B**) in the first direction X and the second direction Y and thus effectively inducing the propagation of the crack (CR of FIG. **9B**) in the third direction Z.

In the dicing region **104**, the multi-layer wires **120** may be formed as a multi-layer dummy wire structure corresponding to metal wires formed in the integrated circuit regions **102**.

Although the multi-layer wires **120** are shown as three layers, they are not limited thereto. For example, the multi-layer wires **120** may be formed as two layers or four or more layers.

The upper material film **130** may be formed on the multi-layer wires **120**. That is, a level of the lowermost surface of the upper material film **130** may be the same as or higher than a level of the uppermost surface of the multi-layer wires **120**.

The upper material film **130** may be formed in a form where a first material film **132**, a second material film **134**, and a third material film **136** are sequentially stacked. Each of the first material film **132**, second material film **134**, and third material film **136** may include an insulating film and may be formed of different materials. In some exemplary embodiments of the disclosure, the first material film **132** may include a silicon oxide, the second material film **134** may include a silicon carbonitride (SiCN), and the third material film **136** may include a silicon nitride. Although the upper material film **130** is shown as three films, it is not limited thereto. For example, unlike shown, the upper material film **130** may be formed as two films or four or more films.

In other exemplary embodiments of the disclosure, the upper material film **130** may be formed to have a structure where a silicon oxide, such as phosphor silicate glass (PSG), boro-phosphor silicate glass (BPSG), undoped silicate glass (USG), tetra ethyl ortho silicate (TEOS), plasma enhanced-TEOS (PE-TEOS), or high density plasma-chemical vapor deposition (HDP-CVD) oxide, and a silicon nitride are stacked alternately.

According to another exemplary embodiment, the upper material film **130** of the dicing region **104** may include a test pattern for testing electrical characteristics of the semiconductor device SD existing in the integrated circuit regions **102**, a redistribution layer for electrical connection between the test patterns, or an alignment key for aligning a mask.

According to an exemplary embodiment of the disclosure, a protective film **140** may be formed to expose the upper material film **130** located in the dicing region **104W** and to cover the upper material film **130** located in the integrated circuit regions **102**. A side wall of the protective film **140** may be an inclined plane. The protective film **140** may include a material film formed of, for example, an organic compound. In some exemplary embodiments of the disclosure, the protective film **140** may include a material film formed of an organic high-polymer material. In other exemplary embodiments of the disclosure, the protective film **140** may include photosensitive polyimide (PSPI) resin. A width **104W** of the dicing region **104** exposed from the protective film **140** may be about 5 μm to about 100 μm . However, a numerical value of the width **104W** is not limited to the above example.

The metal shield layer MS may be formed on the active surface **100F** of the semiconductor substrate **100** across the integrated circuit regions **102** and the dicing region **104**. The metal shield layer MS may prevent a spot generated due to leakage or scattering of laser in a dicing process from spreading to the integrated circuit regions **102**.

A bottom surface of the metal shield layer MS may be arranged to directly contact the active surface **100F**. In other words, the metal shield layer MS may be formed in a middle-end-of-line (MEOL) process. Thus, in the integrated

circuit regions **102** of the semiconductor substrate **100**, the semiconductor device SD may not be arranged in a location where the metal shield layer MS is formed.

In a side cross-section view, the metal shield layer MS may include a first metal shield layer MS1 and a second metal shield layer MS2 with a space region SS of a first interval therebetween in the dicing region **104**. In terms of locations, the first metal shield layer MS1 may be referred to as a left metal shield layer and the second metal shield layer MS2 may be referred to as a right metal shield layer. The first interval of the space region SS may be narrower than the width **104W** of the dicing region **104**.

The first metal shield layer MS1 may have a major axis MS1R in the first direction X that is parallel to the active surface **100F** and a minor axis MS in the third direction Z that is perpendicular to the active surface **100F**. The second metal shield layer MS2 may have a major axis MS2R in the first direction X that is parallel to the active surface **100F** and a minor axis MS2S in the third direction Z that is perpendicular to the active surface **100F**.

Lengths of the major axes MS1R and MS2R may be about 50 μm to about 100 μm , and lengths of the minor axes MS1S and MS2S may be about 0.5 μm to about 1 μm . Describing the above examples as ratios, a ratio of the lengths of the major axes MS1R and MS2R to the lengths of the minor axes MS and MS2S may be about 50:1 to about 200:1.

That is, each of the first metal shield layer MS1 and the second metal shield layer MS2 may be formed as a thin flat plate-type structure to cover the semiconductor substrate **100**. The first metal shield layer MS1 and the second metal shield layer MS2 may substantially have the same shape as each other, but the disclosure is not limited thereto.

In a side cross-section view, a propagation direction of a crack (CR of FIG. 9B) and a major-axial direction of the metal shield layer MS may be perpendicular to each other. That is, the propagation direction of the crack (CR of FIG. 9B) may be the third direction Z, and the major-axial direction of the metal shield layer MS may be the first direction X.

In a plane view, the metal shield layer MS may include single metal in a hollow rectangular or square shape in each of adjacent integrated circuit regions **102** with the dicing region **104** therebetween. To allow the crack (CR of FIG. 9B) to pass through the space region SS, the metal shield layers MS may be arranged to be spaced apart from each other by a distance. In other words, the metal shield layer MS may be arranged to surround a semiconductor chip (**10** of FIG. 10).

A material of the metal shield layer MS may be metal having a melting point of about 600° C. or higher. To prevent the metal shield layer MS from being melted, the metal shield layer MS may include metal having a melting point higher than a temperature of a portion of the semiconductor substrate **100** heated by laser. In some exemplary embodiments of the disclosure, the metal shield layer MS may include aluminum (a melting point of about 660° C.).

Recently, as large volume and high integration of a semiconductor device are required, an area occupied by a dicing region in a semiconductor substrate is reduced. Generally, the semiconductor substrate is mechanically diced using a sawing blade. As such, when mechanical dicing is performed, the risk of damage to integrated circuit regions may increase due to stress applied to the semiconductor substrate during a dicing process.

Hence, a process of dicing a semiconductor substrate by using laser has been performed. However, due to a difference between a density of a modified layer of the semicon-

ductor substrate and a density of a periphery of the modified layer, a part of laser that has to be focused onto an inside of the semiconductor substrate leaks or is scattered, and penetrates a semiconductor device of an integrated circuit region, causing a fault.

Thus, the semiconductor substrate **100** according to the disclosure may prevent the leaking or scattering laser from entering regions where the semiconductor device SD are located by forming the metal shield layer MS across the integrated circuit regions **102** and the dicing region **104** of the semiconductor substrate **100**. In this way, it is possible to prevent a fault such as a function failure occurring in the semiconductor device SD due to leaking or scattering of laser.

As such, a semiconductor chip diced from the semiconductor substrate **100** according to the disclosure has less faults in a dicing process, thereby improving electrical characteristics and production efficiency of the semiconductor chip.

Hereinbelow, a method of dicing the semiconductor substrate **100** including the metal shield layer MS will be described in detail.

FIG. 3 is a perspective view showing a state in which a protective sheet is attached to a semiconductor substrate, according to an exemplary embodiment of the disclosure, and FIG. 4 is a side cross-sectional view corresponding to a line IV-IV' illustrated in FIG. 3, where attachment of a protective sheet to a semiconductor substrate is completed.

Referring to FIGS. 3 and 4, a protective sheet **200** may be attached onto the active surface **100F** of the semiconductor substrate **100**.

The protective sheet **200** may protect the integrated circuit regions **102** during a dicing process of the semiconductor substrate **100**.

The protective sheet **200** may include a polymer sheet based on polyvinylchloride (PVC) and may be attached onto the active surface **100F** using an acrylic resin-based adhesive. The acrylic resin-based adhesive may have a thickness of about 2 μm to about 10 μm , and the protective sheet **200** may have a thickness of about 60 μm to about 200 μm . The protective sheet **200** may have a circular shape having substantially the same diameter as a diameter of the semiconductor substrate **100**.

FIG. 5A is a perspective view showing a state where laser is irradiated into a semiconductor substrate according to an exemplary embodiment of the disclosure, and FIG. 5B is a side cross-sectional view showing a traveling direction of laser.

Referring to FIGS. 5A and 5B, after the protective sheet **200** is attached onto the active surface **100F** of the semiconductor substrate **100**, laser RA of a wavelength having penetrability with respect to the semiconductor substrate **100** may be controlled to have a converging point **150P** inside the semiconductor substrate **100** and thus may be irradiated along the dicing region **104**.

When the laser RA is irradiated to an inside of the semiconductor substrate **100**, a modified layer (**150** of FIG. 6) may be formed along the dicing region **104** inside the semiconductor substrate **100**. The modified layer (**150** of FIG. 6) may be formed using a laser irradiation apparatus **300**.

The laser irradiation apparatus **300** may include a chuck table **310** supporting the semiconductor substrate **100**, a laser irradiation device **320** irradiating the laser RA to the semiconductor substrate **100** arranged on the chuck table **310**, and an imaging device **330** imaging the semiconductor substrate **100** arranged on the chuck table **310**. The chuck

table 310 may support the semiconductor substrate 100 through suction using vacuum pressure and may move in the first direction X and the second direction Y.

The laser irradiation device 320 may be configured to irradiate pulse laser from a condenser 324 mounted on a front end of a housing 322 in a cylindrical shape arranged substantially horizontally. While the condenser 324 irradiates pulse laser of a wavelength having penetrability with respect to the semiconductor substrate 100, the chuck table 310 and the condenser 324 may perform relative movement at a proper speed.

The imaging device 330 mounted on another front end of the housing 322 constituting the laser irradiation device 320 may be a general charge-coupled device (CCD) imaging device that performs imaging by using visible rays. In another exemplary embodiment of the disclosure, the imaging device 330 may include an infrared irradiation device that irradiates infrared rays to the semiconductor substrate 100, an optical system that captures the infrared rays irradiated by the infrared irradiation rays, and an infrared CCD imaging device that outputs an electrical signal corresponding to the infrared rays captured by the optical system.

The laser irradiation device 320 may irradiate the laser RA after being aligned in a laser irradiation location. The converging point 150P of the laser RA may be controlled to be located closer to the active surface 100F than to the inactive surface 100B of the semiconductor substrate 100. That is, the modified layer (150 of FIG. 6) may be located closer to the active surface 100F.

The laser RA emitted from the laser irradiation device 320 may be intensively irradiated such that a part of the semiconductor substrate 100 is heated to a temperature of about 600° C. That is, the part of the semiconductor substrate 100 located in the converging point 150P of the laser RA may be melted by the laser RA.

As such, when the part of the semiconductor substrate 100 is melted, a crystalline state of the part changes, causing unintended leakage or scattering RB of the laser RA. The leakage or scattering RB of the laser RA may cause a spot in a part other than the converging point 150P of the semiconductor substrate 100, and the spot may cause a fault of the semiconductor device.

FIG. 6 is a side cross-sectional view showing a state where irradiation of laser is completed, according to an exemplary embodiment of the disclosure, FIG. 7A, which corresponds to FIG. 2A, is an enlarged plane view of a region A illustrated in FIG. 6, and FIG. 7B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. 7A in a direction X.

Referring to FIGS. 6, 7A and 7B, the modified layer 150 may be located away from the inactive surface 100B of the semiconductor substrate 100 by a first distance D1, and the modified layer 150 may be located closer to the active surface 100F.

Laser may be easily irradiated to a target location through light amplification by stimulated emission radiation. By using properties of the laser, the modified layer 150 may be formed in a target location inside the semiconductor substrate 100. The modified layer 150 may include a crack site where the crack (CR of FIG. 9B) is initiated by an external physical impact.

A first spot 151 and a second spot 153 may be formed due to unintended leakage or scattering of laser. The first spot 151 may indicate a spot formed inside the semiconductor substrate 100, and the second spot 153 may indicate a spot formed inside the metal shield layer MS. The second spot 153 is formed in a location that may substantially affect the

semiconductor device SD, but the semiconductor substrate 100 according to the disclosure may solve such a problem by including the metal shield layer MS.

The modified layer 150 may be located under the semiconductor device layer 110. The modified layer 150 has a constant width in the first direction X, and a virtual line connecting the modified layers 150 in the second direction Y may have a linear lane form.

FIG. 8 is a side cross-sectional view showing a state where a semiconductor substrate according to an exemplary embodiment of the disclosure is polished, FIG. 9A, which corresponds to FIG. 2A, is an enlarged plane view of a region A illustrated in FIG. 8, and FIG. 9B is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. 9A in a direction X.

Referring to FIGS. 8, 9A and 9B, the crack CR may be initiated in the dicing region 104 by polishing the inactive surface 100B of the semiconductor substrate 100.

By polishing the inactive surface 100B of the semiconductor substrate 100 using a polishing device 400, a thickness of the semiconductor substrate 100 may be reduced and the crack CR may propagate from the modified layer 150.

The polishing device 400 may include a chuck table 410 that supports the semiconductor substrate 100 and a grinder 420 that polishes the semiconductor substrate 100 arranged on the chuck table 410. The grinder 420 may move while rotating, and a polishing pad may be attached under the grinder 420.

The semiconductor substrate 100, which is polished, may have a second thickness T2 that is substantially less than the initial first thickness (T1 of FIG. 2B). The second thickness T2 may be about 20 μm to about 50 μm.

By polishing the semiconductor substrate 100 using the polishing device 400, the semiconductor substrate 100, which is polished, having a final thickness may be formed. At the same time, in the dicing region 104, the crack CR may pass through the active surface 100F of the polished semiconductor substrate 100 from the modified layer 150 and propagate in the third direction Z away from the active surface 100F.

According to an exemplary embodiment of the disclosure, laser may be irradiated to the inside of the semiconductor substrate 100 to form the modified layer 150 along the dicing region 104 of the semiconductor substrate 100, and then the inactive surface 100B of the semiconductor substrate 100 may be polished. A polishing process may be a grinding process in a state where physical pressure is applied to the semiconductor substrate 100.

When the polishing process is performed in the state where the physical pressure is applied to the semiconductor substrate 100, the polished semiconductor substrate 100 may be brittle and fractured. The brittle fracture indicates that when a force greater than or equal to an elastic limit is applied to an object, the object is fractured instead of permanently deformed. Thus, during polishing of the inactive surface 100B of the semiconductor substrate 100, the semiconductor substrate 100 that is gradually thinner may be brittle fractured by the crack CR propagating from the modified layer 150. As the crack CR propagating from the modified layer 150 is initiated along the dicing region 104 isolating the integrated circuit regions 102, the integrated circuit regions 102 may be separated into semiconductor chips (10 of FIG. 10) by the brittle fracture of the semiconductor substrate 100. The semiconductor chip (10 of FIG. 10) may be fixed in an original location without leaving the original location, by the protective sheet 200.

11

In other exemplary embodiments of the disclosure, by continuing polishing the inactive surface **100B** of the semiconductor substrate **100**, the modified layer **150** may be completely removed. A diced surface of the semiconductor chip (**10** of FIG. **10**) separated with complete removal of the modified layer **150** may be smoother than a diced surface that is mechanically diced using a sawing blade. Moreover, by completely removing the modified layer **150** in the polishing process, the crack site in the modified layer **150** may be entirely removed, without causing another crack CR.

The crack CR propagating from the modified layer **150** may propagate in the third direction Z. Even when the crack CR partially propagates in the first direction X and/or the second direction Y, the crack CR may not propagate to the integrated circuit regions **102** by being blocked by the plurality of metal vertical structures **122**.

Moreover, by using laser, a cut width of the semiconductor substrate **100** may be reduced. Thus, the width of the dicing region **104** may be reduced as compared to mechanical dicing using a sawing blade, forming more integrated circuit regions **102** on the semiconductor substrate **100**.

FIG. **10** is a side cross-sectional view showing a state where a semiconductor substrate according to an exemplary embodiment of the disclosure is diced into semiconductor chips, FIG. **11A**, which corresponds to FIG. **2A**, is an enlarged plane view of a region A illustrated in FIG. **10**, and FIG. **11B** is a side cross-sectional view corresponding to a surface taken by dicing a center of FIG. **11A** in a direction X.

Referring to FIGS. **10**, **11A** and **11B** together, the semiconductor substrate **100** may be separated into the semiconductor chips **10** by a dicing process.

More specifically, in the semiconductor substrate **100**, the integrated circuit regions **102** may be separated into the semiconductor chips **10** by the crack (CR of FIG. **9B**) in the dicing region **104**. The separated semiconductor chips (**10** of FIG. **10**) may be fixed in an original location without leaving the original location, by the protective sheet **200**.

A method of dicing the semiconductor substrate **100** according to the disclosure may form the metal shield layer MS across the integrated circuit regions **102** and the dicing region **104** of the semiconductor substrate **100**, thereby preventing leaking or scattered laser from affecting the semiconductor device and reducing the width of the dicing region **104**.

In other words, in the method of dicing the semiconductor substrate **100** using laser, an influence of leakage or scattering of the laser may be removed by the metal shield layer MS, thereby preventing a fault such as a function failure of the semiconductor device.

Eventually, a fault of the semiconductor chip **10** diced by the method of dicing the semiconductor substrate **100** may be reduced, and electrical characteristics and productivity of the semiconductor chip **10** may be improved.

FIGS. **12A** and **12B** illustrate a semiconductor substrate according to another exemplary embodiment of the disclosure, in which FIG. **12A** is an enlarged side cross-sectional view of a region A illustrated in FIG. **1** and FIG. **12B** is an enlarged plane view of a region C illustrated in FIG. **1**.

Hereinbelow, many components of a semiconductor substrate **100-1** and materials of the components are substantially the same as or similar to a description made with reference to FIGS. **1**, **2A**, **2B** and **2C**. Thus, for convenience, a description will be described based on differences than the semiconductor substrate (**100** of FIG. **1**) described above.

Referring to FIGS. **12A** and **12B**, the semiconductor substrate **100-1** may include the metal shield layer MS

12

continuing across the adjacent integrated circuit regions **102** with the dicing region **104** therebetween.

The metal shield layer MS may be formed on the active surface **100F** of the semiconductor substrate **100** across the adjacent integrated circuit regions **102** and the dicing region **104**. The metal shield layer MS may prevent a spot generated due to leakage or scattering of laser in a dicing process from spreading to the integrated circuit regions **102**.

A bottom surface of the metal shield layer MS may be arranged to directly contact the active surface **100F**. In other words, the metal shield layer MS may be formed in an MEOL process. Thus, in the integrated circuit regions **102** of the semiconductor substrate **100**, the semiconductor device SD may not be arranged in a location where the metal shield layer MS is formed.

In a side cross-section view, the metal shield layer MS may have a major axis MSR in the first direction X that is parallel to the active surface **100F** and a minor axis MSS in the third direction Z that is perpendicular to the active surface **100F**.

A length of the major axis MSR may be about 100 μm to about 200 μm , and a length of the minor axis MSS may be about 0.5 μm to about 1 μm . Describing the above example as a ratio, a ratio of the length of the major axis MSR to the length of the minor axis MSS may be about 100:1 to about 400:1.

That is, the metal shield layer MS may be formed as a thin flat plate-type structure to cover the semiconductor substrate **100**.

In a side cross-section view, a propagation direction of a crack (CR of FIG. **9B**) and a major-axial direction of the metal shield layer MS may be perpendicular to each other. That is, the propagation direction of the crack (CR of FIG. **9B**) may be the third direction Z, and the major-axial direction of the metal shield layer MS may be the first direction X. In a dicing process, the crack (CR of FIG. **9B**) may penetrate a central portion of the metal shield layer MS located in the dicing region **104**.

In a plane view, the metal shield layer MS may include integrated metal in a hollow lattice shape in each of adjacent integrated circuit regions **102**, covering the active surface **100F** corresponding to the dicing region **104**. The metal shield layer MS may be arranged to allow the crack (CR of FIG. **9B**) to pass through the metal shield layer MS. In other words, the metal shield layer MS may be arranged to surround a semiconductor chip (**10** of FIG. **10**).

A material of the metal shield layer MS may be metal having a melting point of about 600° C. or higher. To prevent the metal shield layer MS from being melted, the metal shield layer MS may include metal having a melting point higher than a temperature of a portion of the semiconductor substrate **100** heated by laser. In some exemplary embodiments of the disclosure, the metal shield layer MS may include aluminum (a melting point of about 660° C.).

FIG. **13** illustrates a semiconductor substrate according to another exemplary embodiment of the disclosure, in which FIG. **13** is an enlarged side cross-sectional view of a region A illustrated in FIG. **1**;

Hereinbelow, many components of a semiconductor substrate **100-2** and materials of the components are substantially the same as or similar to a description made with reference to FIGS. **1** through **2C**. Thus, for convenience, a description will be described based on differences than the semiconductor substrate (**100** of FIG. **1**) described above.

13

Referring to FIG. 13, a semiconductor substrate **100-2** may include the integrated circuit regions **102** and the dicing region **104**, without including a plurality of metal vertical structures.

The multi-layer wires **120** may be formed from a top surface of the semiconductor device layer **110** to a bottom surface of an upper material film **130**. The multi-layer wires **120** may include an inter-layer insulating film **124** and a metal wire **126** that are arranged alternately.

The metal wire **126** may include a conductive material including at least one of aluminum (Al), copper (Cu), nickel (Ni), tungsten (W), platinum (Pt), and gold (Au).

The inter-layer insulating film **124** may include a low dielectric material. The low dielectric material, which has a lower dielectric constant than that of a silicon oxide, may be useful for high integration and high speed of the semiconductor device SD when the low dielectric material is used as the inter-layer insulating film **124** in the semiconductor device SD.

In some exemplary embodiments of the disclosure, the inter-layer insulating film **124** may be formed to have a structure in which a first inter-layer insulating film, a second inter-layer insulating film, and a third inter-layer insulating film are sequentially stacked with the metal wire **126** thereamong. However, the number of films of the inter-layer insulating film **124** is not limited to the above-described example. The inter-layer insulating film **124** may be formed to fill the peripheries of the plurality of metal vertical structures **122** and the metal wire **126**.

The metal shield layer MS may be formed on the active surface **100F** of the semiconductor substrate **100** across the integrated circuit regions **102** and the dicing region **104**. The metal shield layer MS may prevent a spot generated due to leakage or scattering of laser in a dicing process from spreading to the integrated circuit regions **102**.

In a side cross-section view, the metal shield layer MS may include a first metal shield layer MS1 and a second metal shield layer MS2 with a space region SS of a first interval therebetween in the dicing region **104**. In terms of locations, the first metal shield layer MS1 may be referred to as a left metal shield layer and the second metal shield layer MS2 may be referred to as a right metal shield layer. The first interval of the space region SS may be narrower than the width **104W** of the dicing region **104**.

FIG. 14 is a side cross-sectional view illustrating a semiconductor package including semiconductor chips diced from a semiconductor substrate according to an exemplary embodiment of the disclosure, FIG. 15A is an enlarged side cross-sectional view of a portion D illustrated in FIG. 14 according to an exemplary embodiment of the disclosure, and FIG. 15B is an enlarged side cross-sectional view of a portion D illustrated in FIG. 14 according to an exemplary embodiment of the disclosure.

Referring to FIG. 14, a semiconductor package **1000** may include a package substrate **1010**, the plurality of semiconductor chips **10**, a connection member **1020**, and a molding member **1030**.

The package substrate **1010** as a support substrate may include a body **1011**. According to exemplary embodiment, the package substrate **1010** may further include a lower protective layer, and an upper protective layer. The package substrate **1010** may be formed based on a printed circuit board (PCB), a wafer substrate, a ceramic substrate, a glass substrate, an interposer substrate, etc. In an exemplary embodiment of the disclosure, the package substrate **1010** may be a PCB. The package substrate **1010** is not limited to a PCB.

14

According to an embodiment, a wire **1016** may be formed on the package substrate **1010**, and the wire **1016** may be electrically connected to the semiconductor chip **10** through the connection member **1020** connected to the upper electrode pad **1012** on a top surface of the package substrate **1010**. An external connection terminal **1040** may be arranged on the lower electrode pad **1014** on a bottom surface of the package substrate **1010**. The package substrate **1010** may be mounted in electrical connection with a module substrate or a system board of an electronic product through the external connection terminal **1040**.

The multi-layer or single-layer wire, for example, the wire **1016**, may be formed in the body **1011**, and the semiconductor chip **10** may be electrically connected with the external connection terminal **1040** through the wire **1016**. The lower protective layer and the upper protective layer may function to protect the body **1011**, and may be formed as, for example, solder resist.

When the package substrate **1010** is a PCB, the body **1011** may be generally implemented by compressing a high-polymer material such as thermoset resin, epoxy-based resin such as flame retardant **4** (FR-4), bismaleimide triazine (BT), an Ajinomoto build up film (ABF), etc., phenol resin, etc., to form a thin film, applying copper foil to opposite sides, and forming the wire **1016** that is a delivery path of an electrical signal through patterning. Except for parts connected with a terminal, e.g., the upper electrode pad **1012** and the lower electrode pad **1014**, solder resist may be coated onto a bottom surface and a top surface of the body **1011**, thus implementing the lower protective layer and the upper protective layer.

Meanwhile, the PCB may be divided into a single layer PCB in which the wire **1016** is formed only on a single side, and a double layer PCB in which the wire **1016** is formed on opposite sides. The number of layers of the copper foil may be three or more by using an insulator called prepreg, and three or more wires **1016** may be formed based on the number of formed copper foils, thus implementing a PCB having a multi-layer structure. However, the package substrate **1010** is not limited to the above-described structure or material of the PCB.

The plurality of semiconductor chips **10** may be electrically connected to the package substrate **1010** by the connection member **1020**. The connection member **1020** may electrically connect the package substrate **1010** and the plurality of semiconductor chips **10** by electrically connecting the upper electrode pad **1012** of the package substrate **1010** with the connection pad **12** of the semiconductor chip **10**. In some exemplary embodiments of the disclosure, the connection member **1020** may be a bonding wire.

The connection member **1020** may be used to electrically connect the semiconductor chip with the package substrate **1010**. Through the connection member **1020**, at least one of a control signal, a power signal, and a ground signal for an operation of the semiconductor chip **10** may be provided from outside, a data signal to be stored in the semiconductor chip **10** may be provided from outside, or data stored in the semiconductor chip **10** may be provided to outside.

The molding member **1030** may protect the plurality of semiconductor chips **10** from an external environment by surrounding the plurality of semiconductor chips **10**. The molding member **1030** may be used to inject a proper amount of molding resin onto the package substrate **1010** through an injection process and form an appearance of the semiconductor package **1000** through a hardening process. Depending on a need, in a pressing process, the appearance of the semiconductor package **1000** may be formed by

15

applying pressure to the molding resin. Herein, a processing condition such as a delay time between injection of the molding resin and the pressing, the amount of injection molding resin, and the pressing temperature/pressure, etc., may be set based on physical properties such as viscosity of molding resin.

A side surface and a top surface of the molding member **1030** may have a right-angled shape. In a process of manufacturing the semiconductor package **1000** by dicing the package substrate **1010** along a dicing line, the side surface and the top surface of the molding member **1030** may generally have a right-angled shape. Although not shown, a marking pattern including information of the semiconductor chip **10**, e.g., a barcode, a number, a character, a symbol, etc., may be formed in a part of the side surface of the semiconductor package **1000**.

In some exemplary embodiments of the disclosure, the molding resin may include epoxy-group molding resin, polyimide-group molding resin, etc. The molding member **1030** may include, for example, an epoxy molding compound (EMC).

The connection pad **12** may be arranged on a semiconductor device layer, and may be electrically connected with a wire layer inside the semiconductor device layer. The wire layer may be electrically connected with the connection member **1020** through the connection pad **12**. The connection pad **12** may include at least one of aluminum (Al), copper (Cu), nickel (Ni), tungsten (W), platinum (Pt), and gold (Au).

On the semiconductor device layer may be formed a passivation layer for protecting the semiconductor device layer, the wire layer, and other structures from an external shock or moisture. The passivation layer may expose at least a part of the top surface of the connection pad **12**.

The plurality of semiconductor chips **10** constituting the semiconductor package **1000** may have a stacked structure. When eight semiconductor chips **10** are stacked as shown in the figures, four semiconductor chips **10** may form a group and the semiconductor package **1000** may include two groups.

When the four semiconductor chips **10** of a first group are stacked from the bottom layer to the top layer one by one, the semiconductor chips **10** may be stepwisely moved for arrangement in the first direction X to expose the connection pad **12** arranged in each semiconductor chip **10**.

The four semiconductor chips **10** of a second group may be stacked on the first group. The four semiconductor chips **10** of the second group may be stepwisely moved for arrangement in the first direction X opposite to a direction in which the first group is moved.

Each semiconductor chip **10** constituting the semiconductor package **1000** may include a semiconductor chip diced from the semiconductor substrate according to the disclosure. In some exemplary embodiments of the disclosure, a semiconductor chip diced from the semiconductor substrate **100** described above may be included as shown in FIG. **15A**. In other exemplary embodiments of the disclosure, a semiconductor chip diced from the semiconductor substrate **100-1** described above may be included as shown in FIG. **15B**.

FIG. **16** is a structural diagram showing a system of a semiconductor package including semiconductor chips diced from a semiconductor substrate according to an exemplary embodiment of the disclosure.

Referring to FIG. **16**, a system **1100** may include a controller **1110**, an input/output device **1120**, a memory **1130**, an interface **1140**, and a bus **1150**.

16

The system **1100** may be a mobile system or a system that transmits or receives information. In some exemplary embodiments of the disclosure, the mobile system may be a portable computer, a web tablet, a mobile phone, a digital music player, or a memory card.

The controller **1110** is intended to control an execution program in the system **1100**, and may include a microprocessor, a digital signal processor, a microcontroller, or a similar device.

The input/output device **1120** may be used to input or output data of the system **1100**. The system **1100** may be connected with an external device, e.g., a personal computer (PC) or a network, or exchange data with an external device by using the input/output device **1120**. The input/output device **1120** may be, for example, a touch pad, a keyboard, or a display device.

The memory **1130** may store data for an operation of the controller **1110** or data processed in the controller **1110**. The memory **1130** may include a semiconductor chip diced from the semiconductor substrate according to the disclosure.

The interface **1140** may be a data transmission path between the system **1100** and the external device. The controller **1110**, the input/output device **1120**, the memory **1130**, the interface **1140** may communicate with one another through a bus **1150**.

While one or more aspects of the disclosure has been particularly shown and described with reference to exemplary embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

What is claimed is:

1. A method of dicing a semiconductor wafer, the method comprising:

providing a semiconductor substrate having a plurality of integrated circuit regions on an active surface of the semiconductor substrate, a dicing region provided between adjacent integrated circuit regions of the plurality of integrated circuit regions, and a metal shield layer provided on the active surface across at least a portion of the adjacent integrated circuit regions and the dicing region;

forming a modified layer by irradiating laser to an inside of the semiconductor substrate along the dicing region; propagating a crack from the modified layer in a direction perpendicular to a major-axial direction of the metal shield layer by polishing an inactive surface opposing the active surface of the semiconductor substrate; and forming semiconductor chips by separating the adjacent integrated circuit regions, respectively, based on the crack propagating from the modified layer,

wherein the method comprises forming a plurality of vertical metal structures directly on the metal shield layer, the plurality of vertical metal structures extending in a minor-axial direction that is perpendicular to the active surface,

wherein, prior to the dicing, the metal shield layer is a continuous thin flat plate-type structure having a bottom side surface in direct contact with an upper surface of the semiconductor substrate, and all other side surfaces in direct contact with a semiconductor device layer formed on the semiconductor substrate, except for a portion of one of the side surface contacting the plurality of vertical metal structures, and

wherein the plurality of vertical metal structures continuously extend in the minor-axial direction through a plurality of dielectric layers and metal wiring layers.

17

2. The method of claim 1, wherein,
the forming of the modified layer by irradiating the laser
comprises preventing a spot generated due to leakage
or scattering of the laser from spreading to the adjacent
integrated circuit regions using the metal shield layer. 5

3. The method of claim 1, wherein,
in the providing of the semiconductor substrate, a bottom
surface of the metal shield layer is arranged to directly
contact the active surface.

4. The method of claim 1, wherein 10
the adjacent integrated circuit regions include a first
integrated circuit region and a second integrated circuit
region that are adjacent to each other,
the metal shield layer comprises a first metal shield layer
corresponding to the first integrated circuit region and 15
a second metal shield layer corresponding to the second
integrated circuit region,
the first metal shield layer and the second metal shield
layer are spaced apart from each other by a first
interval, and 20
the first interval is less than a width of the dicing region.

5. The method of claim 4, wherein
each of the first metal shield layer and the second metal
shield layer includes a major axis that is parallel to the
active surface and a minor axis that is perpendicular to 25
the active surface, and
a ratio of a length of each of the first metal shield layer and
the second metal shield layer in the major axis to a
length of each of the first metal shield layer and the
second metal shield layer in the minor axis is about 30
50:1 to about 200:1.

6. The method of claim 1, wherein,
in a planar view, the metal shield layer is continuous
across the adjacent integrated circuit regions with the
dicing region therebetween, and 35
the crack penetrates the metal shield layer located in the
dicing region.

7. The method of claim 6, wherein
the metal shield layer covers the active surface corre-
sponding to the dicing region. 40

8. The method of claim 1, wherein,
in the providing of the semiconductor substrate, the metal
shield layer includes single metal, and
a melting point of material forming the metal shield layer
is higher than about 600° C. 45

9. The method of claim 1, wherein
the semiconductor substrate is provided to include a
plurality of metal vertical structures for inducing propa-
gation of the crack are arranged on the metal shield
layer. 50

10. The method of claim 9, wherein,
in a planar view, the direction of the crack and a major-
axial direction of each of the plurality of metal vertical
structures are parallel to each other.

11. The method of claim 1, wherein the plurality of 55
dielectric layers and metal wiring layers form a stacked layer
including a plurality of inter-layer insulating film and a
plurality of metal wires alternatively arranged.

12. A method of dicing a semiconductor wafer, the 60
method comprising:
providing a semiconductor substrate having a plurality of
integrated circuit regions on an active surface of the
semiconductor substrate, a dicing region provided
between adjacent integrated circuit regions of the plu-
rality of integrated circuit regions, and a metal shield

18

layer formed on the active surface across at least a
portion of the integrated circuit regions and the dicing
region;
forming a modified layer by irradiating laser to an inside
of the semiconductor substrate along the dicing region;
propagating a crack from the modified layer in a direction
perpendicular to a major-axial direction of the metal
shield layer by polishing an inactive surface opposing
the active surface of the semiconductor substrate; and
forming semiconductor chips by separating the adjacent
integrated circuit regions, respectively, based on the
crack propagating from the modified layer,
wherein, in a cross-sectional view, the metal shield layer
includes a first metal shield layer and a second metal
shield layer with a space region therebetween in a
location where the crack propagates,
each of the first metal shield layer and the second metal
shield layer includes a major axis that is parallel to the
active surface and a minor axis that is perpendicular to
the active surface in the cross-sectional view, and
a length of the major axis is about 50 μm to about 100 μm ,
and a length of the minor axis is about 0.5 μm to about
1 μm ,
wherein the method comprises forming a plurality of
vertical metal structures directly on the metal shield
layer, the plurality of vertical metal structures extend-
ing in a minor-axial direction that is perpendicular to
the active surface, and
wherein, prior to the dicing, the metal shield layer is a
continuous thin flat plate-type structure having a bot-
tom side surface in direct contact with an upper surface
of the semiconductor substrate, and all other side
surfaces in direct contact with a semiconductor device
layer formed on the semiconductor substrate, except for
a portion of one of the side surface contacting the
plurality of vertical metal structures, and
wherein the plurality of vertical metal structures continu-
ously extend in the minor-axial direction through a
plurality of dielectric layers and metal wiring layers.

13. The method of claim 12, wherein,
the forming of the modified layer by irradiating the laser
comprises preventing a spot generated due to leakage
or scattering of the laser from spreading to the adjacent
integrated circuit regions using the first metal shield
layer and the second metal shield layer.

14. The method of claim 12, wherein,
the adjacent integrated circuit regions include a first
integrated circuit region and a second integrated circuit
region that are adjacent to each other, and
in the forming of the semiconductor chips, in a plane
view, the first metal shield layer has a rectangular shape
arranged along a circumference of the first integrated
circuit region, and the second metal shield layer has a
rectangular shape arranged along a circumference of
the second integrated circuit region that is adjacent to
the first integrated circuit region.

15. The method of claim 12, wherein,
in the providing of the semiconductor substrate, the first
metal shield layer and the second metal shield layer
have substantially same shape and same material as
each other.

16. The method of claim 15, wherein
the first metal shield layer and the second metal shield
layer include aluminum.

* * * * *